

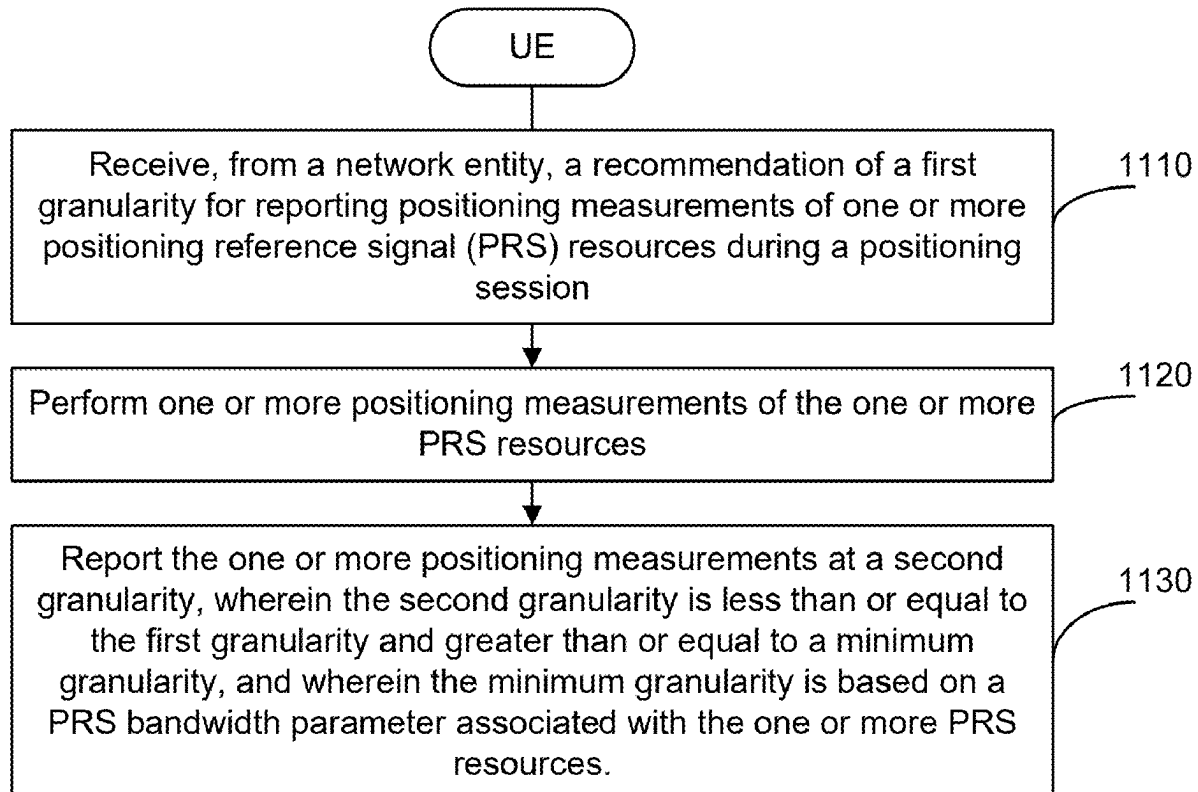


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(19) **United States**(12) **Patent Application Publication**
MANOLAKOS et al.(10) **Pub. No.: US 2025/0287350 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **REPORTING GRANULARITY AND
MEASUREMENT PERIOD FOR
POSITIONING REFERENCE SIGNAL (PRS)
MEASUREMENTS**(71) Applicant: **QUALCOMM Incorporated**, San
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OPSHAUG, Redwood City, CA (US)(21) Appl. No.: **19/216,527**(22) Filed: **May 22, 2025****Related U.S. Application Data**(62) Division of application No. 17/314,258, filed on May
7, 2021, now Pat. No. 12,349,103.(60) Provisional application No. 63/025,510, filed on May
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H04W 24/10 (2009.01)
(52) **U.S. Cl.**
CPC **H04W 64/006** (2013.01); **H04L 5/0051**
(2013.01); **H04W 24/10** (2013.01)(57) **ABSTRACT**

Disclosed are techniques for wireless positioning. In an aspect, a user equipment (UE) receives, from a network entity, a configuration of one or more positioning reference signal (PRS) resources to measure during a positioning session, the one or more PRS resources having a PRS periodicity T_{PRS} and a PRS occasion length L_{PRS} , and measures the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of 'T' milliseconds, the PRS periodicity T_{PRS} , and a measurement gap periodicity of at least one measurement gap.

1100



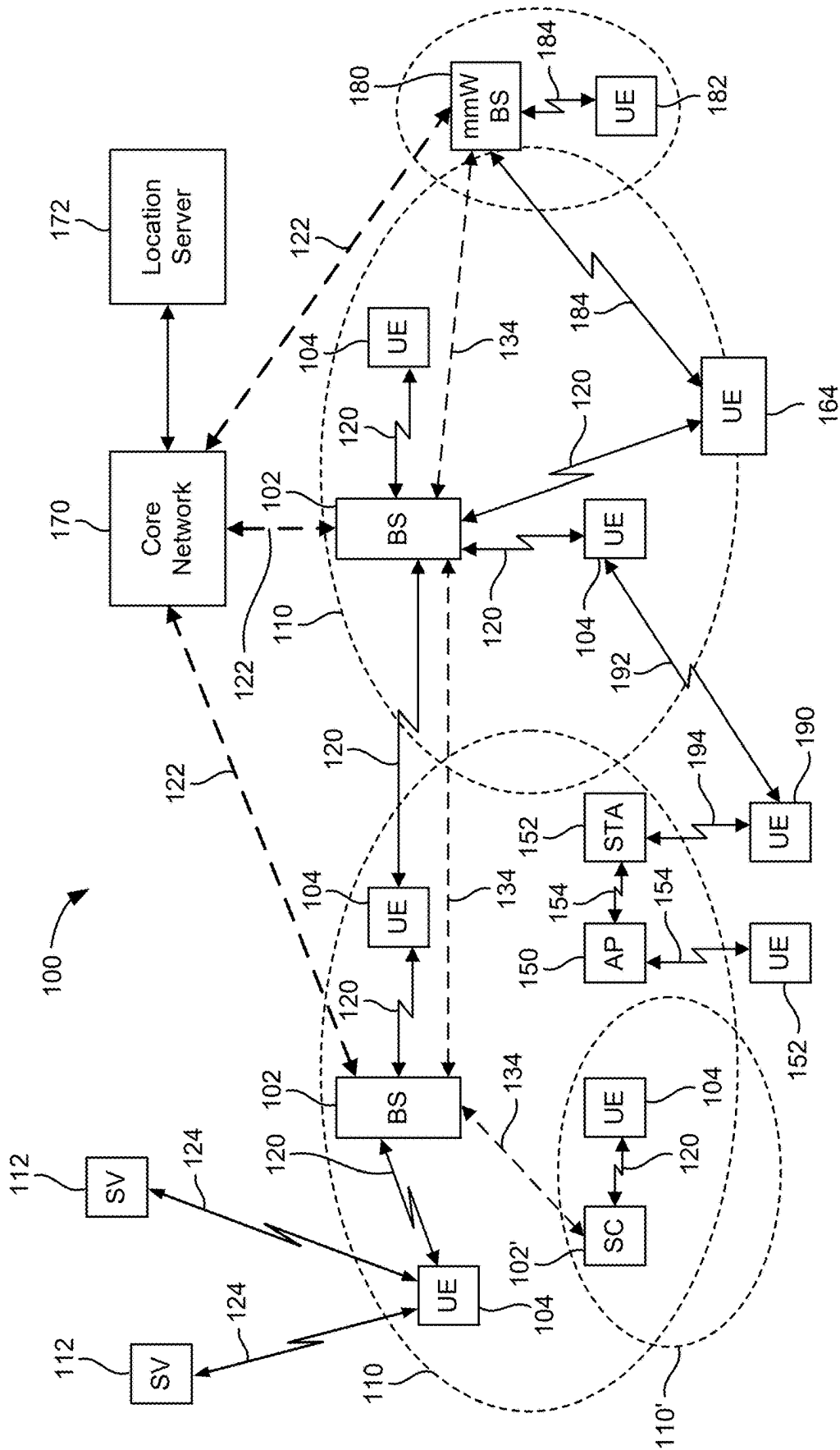


FIG. 1

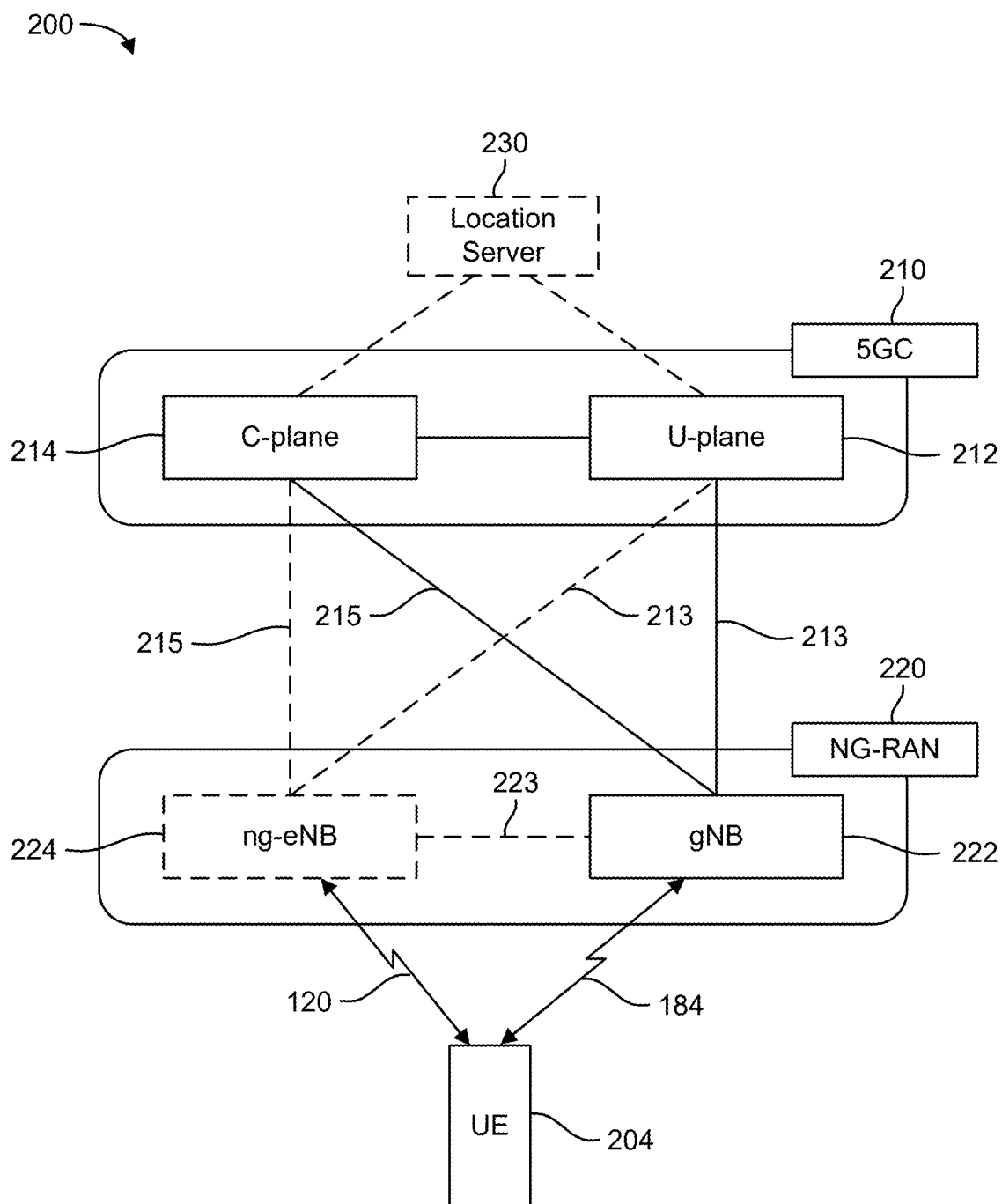


FIG. 2A

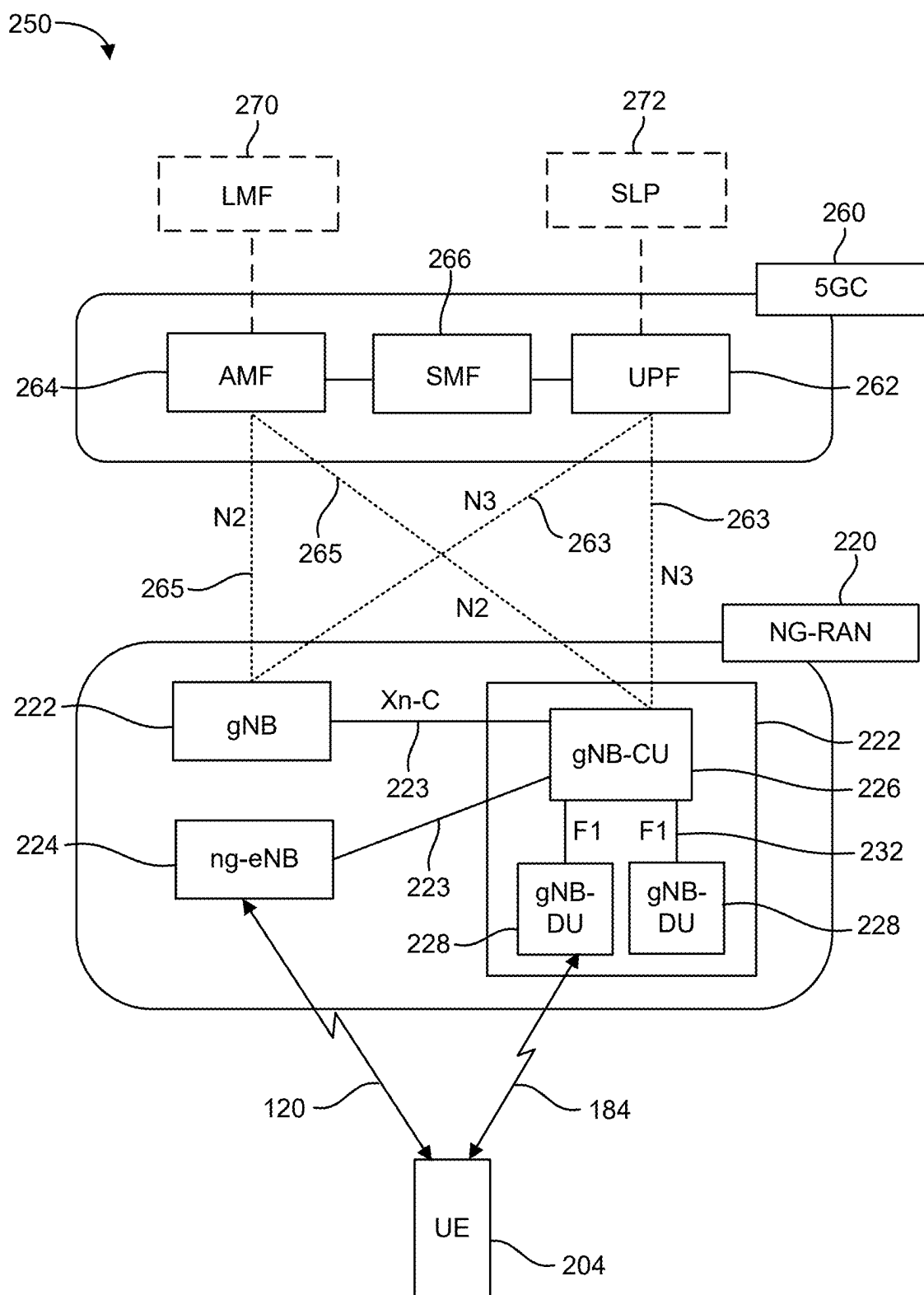


FIG. 2B

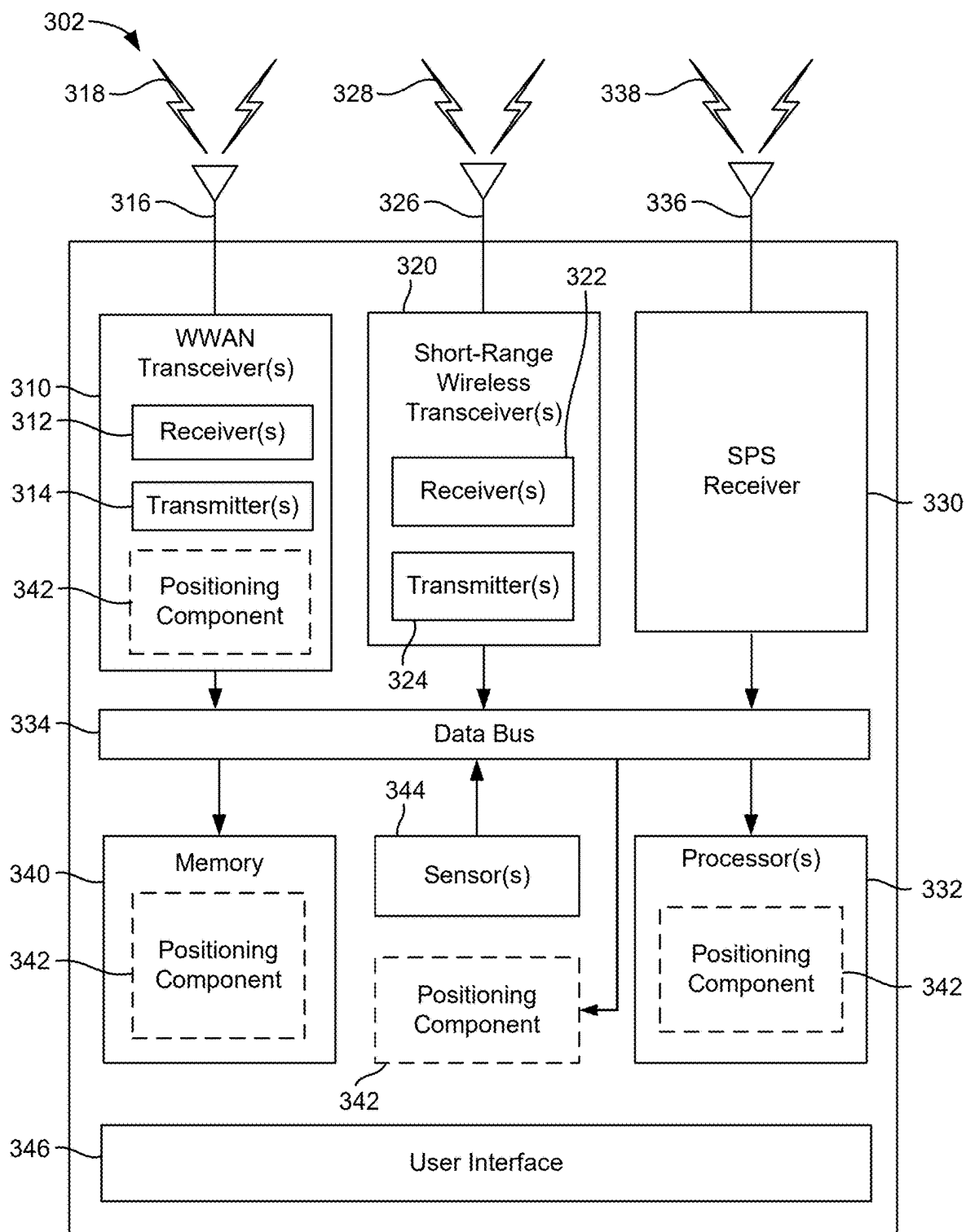


FIG. 3A

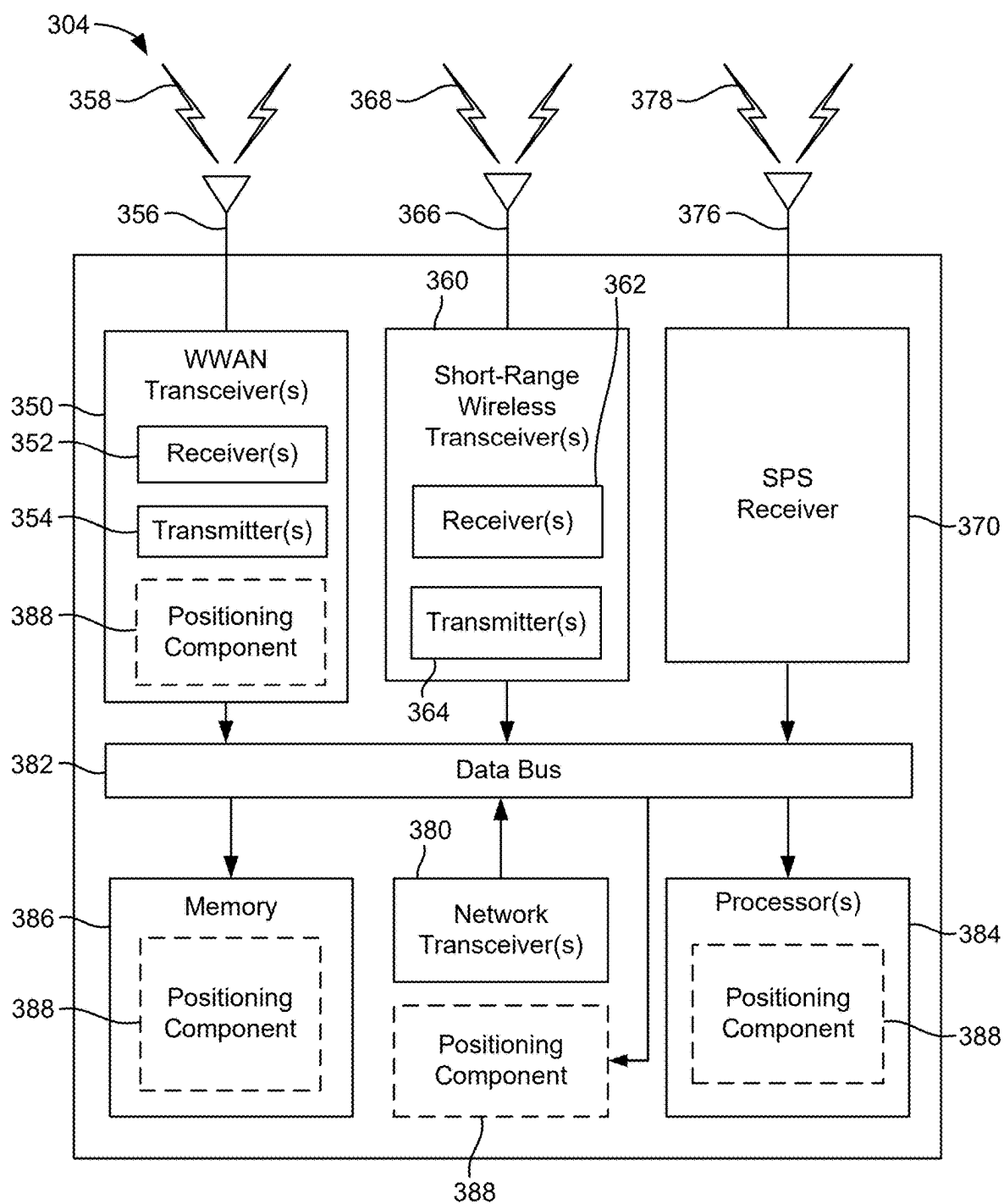


FIG. 3B

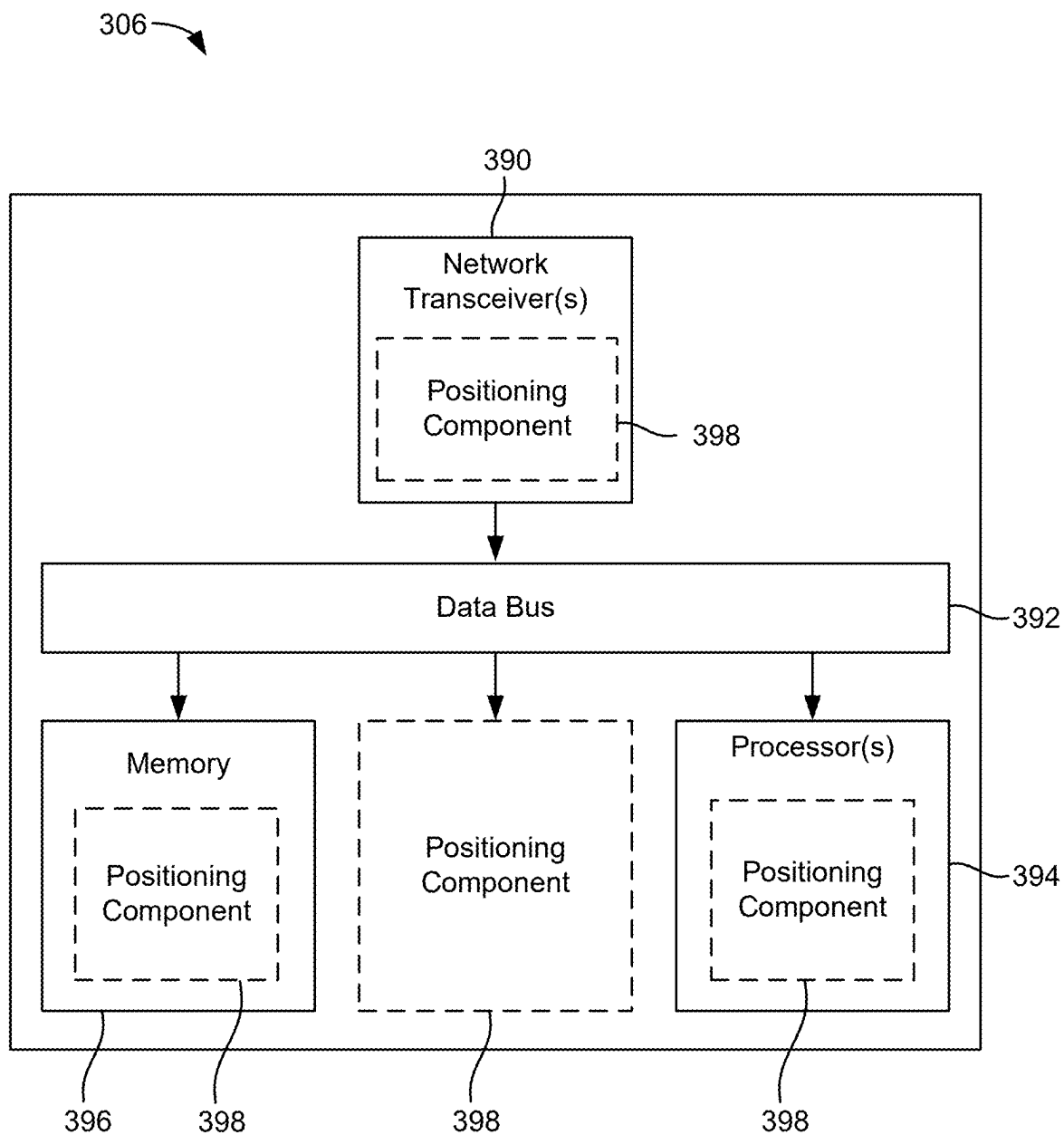


FIG. 3C

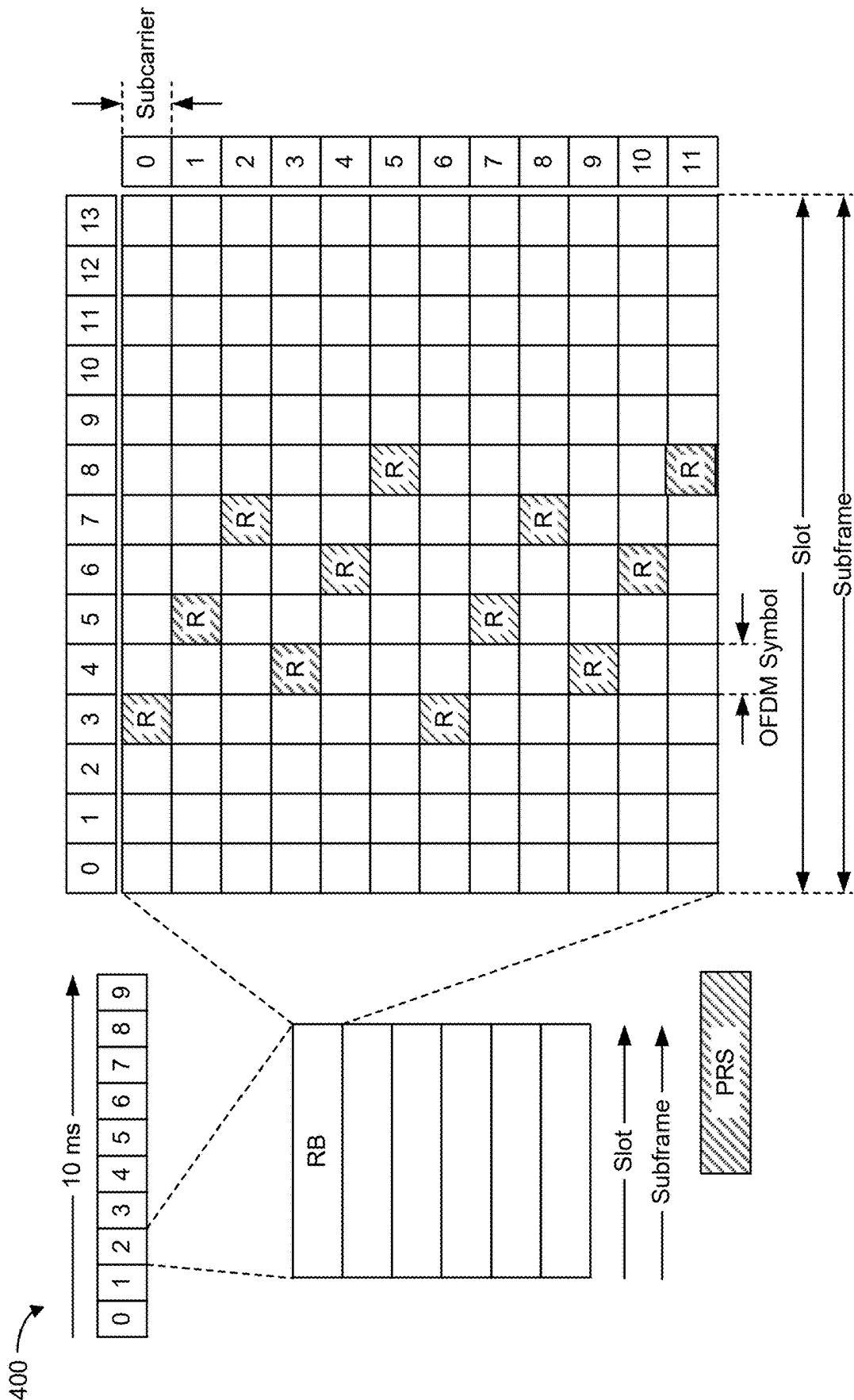


FIG. 4A

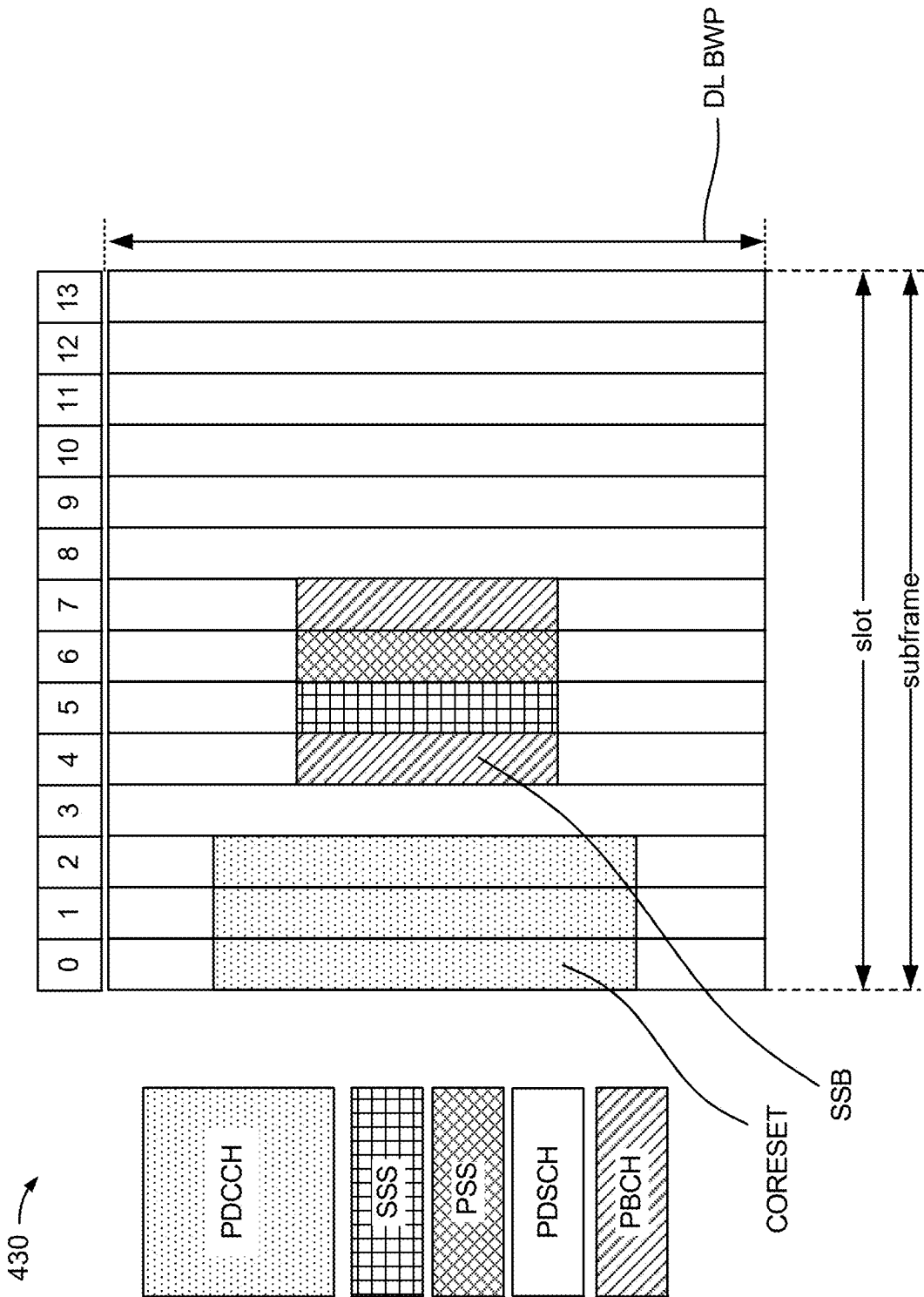


FIG. 4B

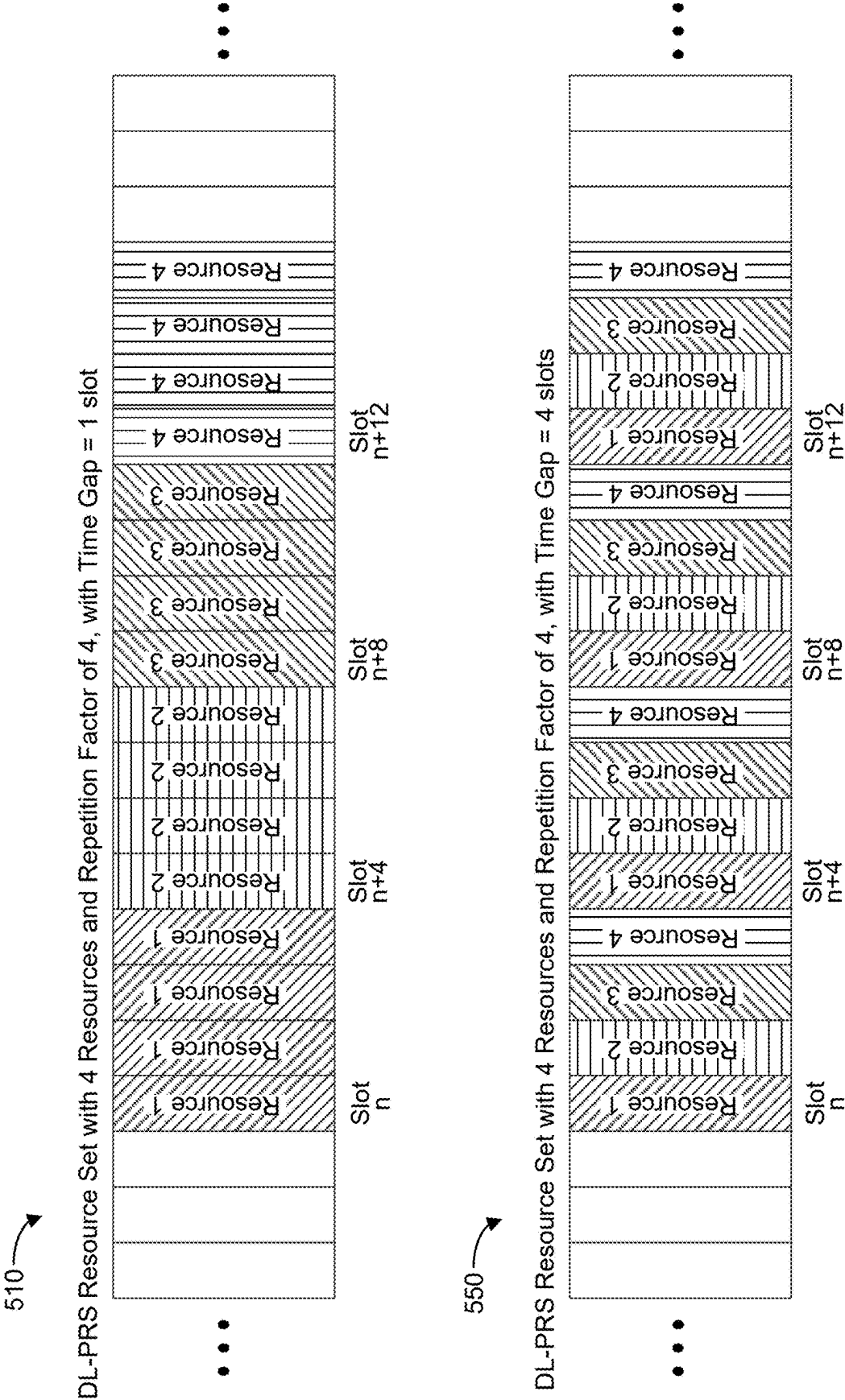


FIG. 5



FIG. 6

710 → NR-Multi-RTT-MeasElement-r16 ::= SEQUENCE {
 trp-ID-r16
 nr-DL-PRS-ResourceId-r16 TRP-ID-r16, OPTIONAL,
 nr-DL-PRS-ResourceSetId-r16 NR-DL-PRS-ResourceSetId-r16 OPTIONAL,
 nr-UE-RxTxTimeDiff-r16 INTEGER OPTIONAL,
 nr-AdditionalPathList-r16 NR-AdditionalPathList-r16 OPTIONAL,
 nr-TimeStamp-r16 NR-TimeStamp-r16,
 nr-TimingMeasQuality-r16 NR-TimingMeasQuality-r16,
 nr-PRS-RSRP-Result-r16 INTEGER OPTIONAL,
 nr-Multi-RTT-AdditionalMeasurements-r16
 NR-Multi-RTT-AdditionalMeasurements-r16 OPTIONAL,
 ...
 }

720 → NR-Multi-RTT-AdditionalMeasurements-r16 ::= SEQUENCE (SIZE (1..3)) OF
 NR-Multi-RTT-AdditionalMeasurementElement-r16

730 → NR-Multi-RTT-AdditionalMeasurementElement-r16 ::= SEQUENCE {
 nr-DL-PRS-ResourceId-r16 NR-DL-PRS-ResourceId-r16 OPTIONAL,
 nr-DL-PRS-ResourceSetId-r16 NR-DL-PRS-ResourceSetId-r16 OPTIONAL,
 nr-PRS-RSRP-ResultDiff-r16 INTEGER OPTIONAL,
 nr-UE-RxTxTimeDiffAdditional-r16 INTEGER OPTIONAL,
 nr-AdditionalPathList-r16 NR-AdditionalPathList-r16 OPTIONAL,
 nr-TimeStamp-r16 NR-TimeStamp-r16,
 ...
 }

FIG. 7

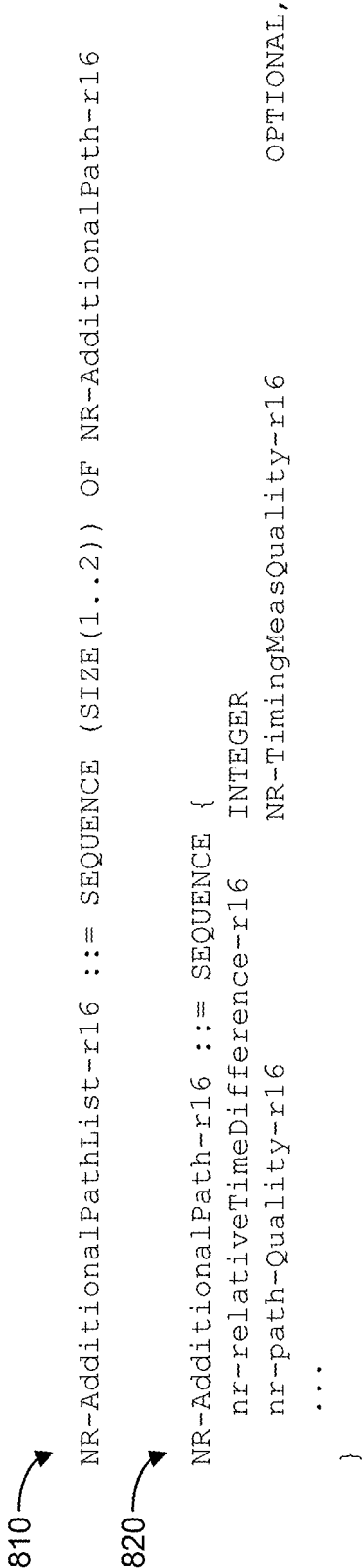


FIG. 8

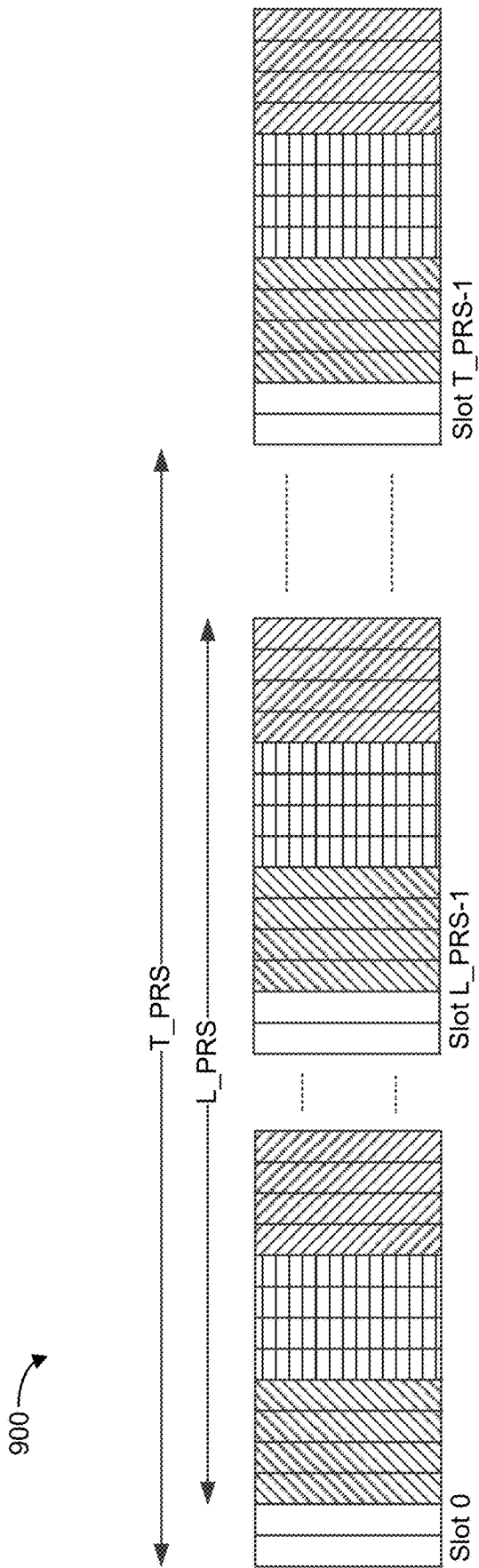


FIG. 9

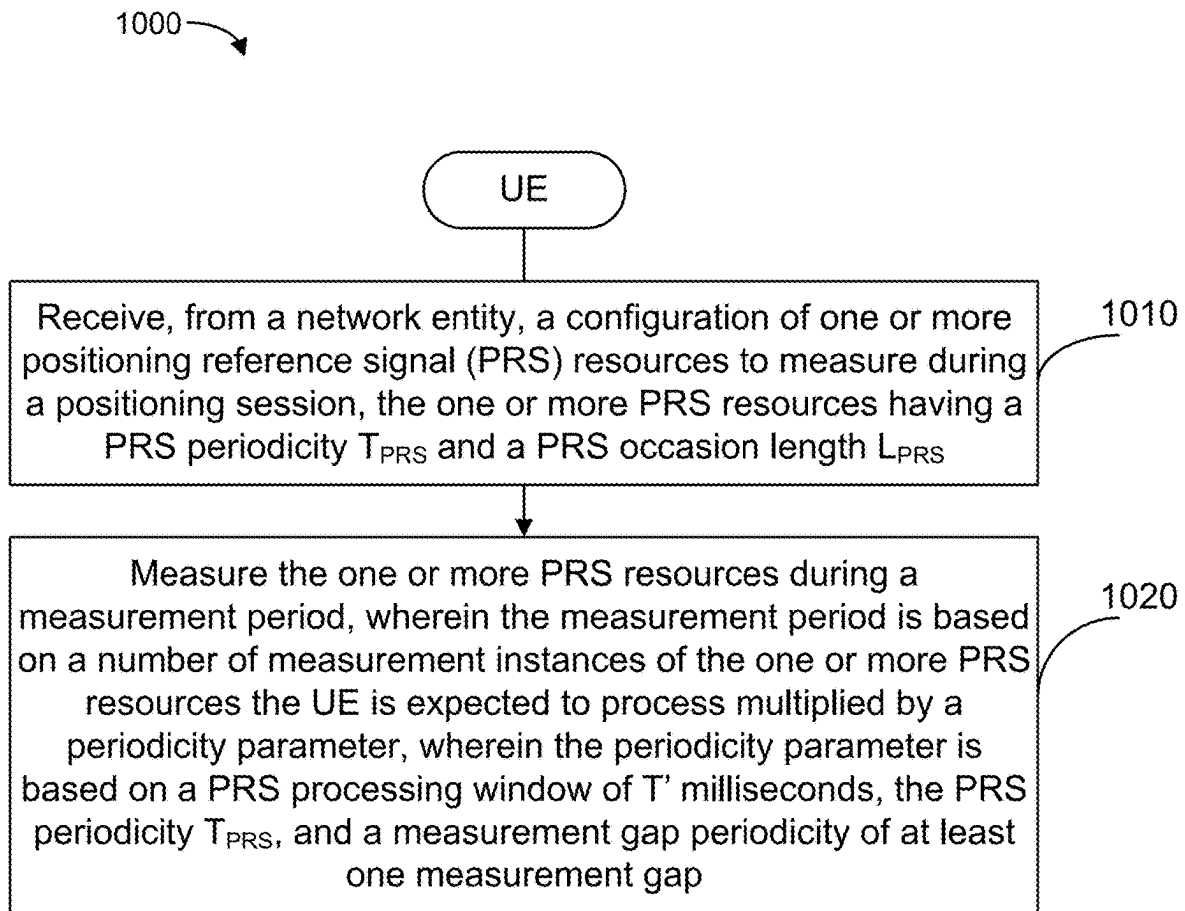


FIG. 10

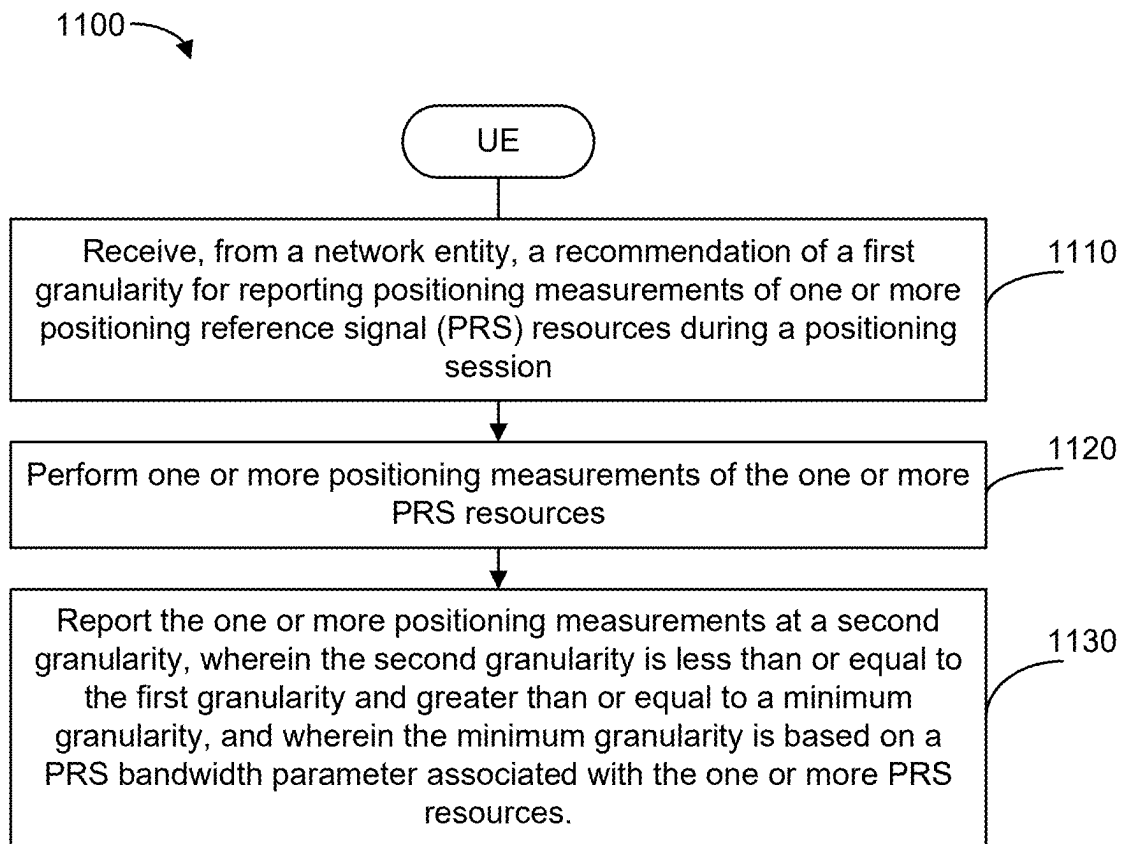


FIG. 11

**REPORTING GRANULARITY AND
MEASUREMENT PERIOD FOR
POSITIONING REFERENCE SIGNAL (PRS)
MEASUREMENTS**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] The present application for patent is a divisional of U.S. application Ser. No. 17/314,258, entitled “REPORTING GRANULARITY AND MEASUREMENT PERIOD FOR POSITIONING REFERENCE SIGNAL (PRS) MEASUREMENTS,” filed May 7, 2021, which claims the benefit of U.S. Provisional Application No. 63/025,510, entitled “REPORTING GRANULARITY AND MEASUREMENT PERIOD FOR POSITIONING REFERENCE SIGNAL (PRS) MEASUREMENTS,” filed May 15, 2020, each of which is assigned to the assignee hereof, and expressly incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] Aspects of the disclosure relate generally to wireless communications.

2. Description of the Related Art

[0003] Wireless communication systems have developed through various generations, including a first-generation analog wireless phone service (1G), a second-generation (2G) digital wireless phone service (including interim 2.5G and 2.75G networks), a third-generation (3G) high speed data, Internet-capable wireless service and a fourth-generation (4G) service (e.g., Long Term Evolution (LTE) or WiMax). There are presently many different types of wireless communication systems in use, including cellular and personal communications service (PCS) systems. Examples of known cellular systems include the cellular analog advanced mobile phone system (AMPS), and digital cellular systems based on code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), the Global System for Mobile communications (GSM), etc.

[0004] A fifth generation (5G) wireless standard, referred to as New Radio (NR), calls for higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard, according to the Next Generation Mobile Networks Alliance, is designed to provide data rates of several tens of megabits per second to each of tens of thousands of users, with 1 gigabit per second to tens of workers on an office floor. Several hundreds of thousands of simultaneous connections should be supported in order to support large sensor deployments. Consequently, the spectral efficiency of 5G mobile communications should be significantly enhanced compared to the current 4G standard. Furthermore, signaling efficiencies should be enhanced and latency should be substantially reduced compared to current standards.

SUMMARY

[0005] The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the

following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

[0006] In an aspect, a method of wireless positioning performed by a user equipment (UE) includes receiving, from a network entity, a configuration of one or more positioning reference signal (PRS) resources to measure during a positioning session, the one or more PRS resources having a PRS periodicity TPRS and a PRS occasion length LPRS; and measuring the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of ‘T’ milliseconds, the PRS periodicity TPRS, and a measurement gap periodicity of at least one measurement gap.

[0007] In an aspect, a method of wireless positioning performed by a user equipment (UE) includes receiving, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session; performing one or more positioning measurements of the one or more PRS resources; and reporting the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

[0008] In an aspect, a user equipment (UE) includes a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: receive, via the at least one transceiver, from a network entity, a configuration of one or more positioning reference signal (PRS) resources to measure during a positioning session, the one or more PRS resources having a PRS periodicity TPRS and a PRS occasion length LPRS; and measure the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of ‘T’ milliseconds, the PRS periodicity TPRS, and a measurement gap periodicity of at least one measurement gap.

[0009] In an aspect, a user equipment (UE) includes a memory; at least one transceiver; and at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to: receive, via the at least one transceiver, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session; perform one or more positioning measurements of the one or more PRS resources; and report the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein

the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

[0010] In an aspect, a user equipment (UE) includes means for receiving, from a network entity, a configuration of one or more positioning reference signal (PRS) resources to measure during a positioning session, the one or more PRS resources having a PRS periodicity TPRS and a PRS occasion length LPRS; and means for measuring the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of ‘T’ milliseconds, the PRS periodicity TPRS, and a measurement gap periodicity of at least one measurement gap.

[0011] In an aspect, a user equipment (UE) includes means for receiving, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session; means for performing one or more positioning measurements of the one or more PRS resources; and means for reporting the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

[0012] In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: receive, from a network entity, a configuration of one or more positioning reference signal (PRS) resources to measure during a positioning session, the one or more PRS resources having a PRS periodicity TPRS and a PRS occasion length LPRS; and measure the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of ‘T’ milliseconds, the PRS periodicity TPRS, and a measurement gap periodicity of at least one measurement gap.

[0013] In an aspect, a non-transitory computer-readable medium stores computer-executable instructions that, when executed by a user equipment (UE), cause the UE to: receive, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session; perform one or more positioning measurements of the one or more PRS resources; and report the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

[0014] Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The accompanying drawings are presented to aid in the description of various aspects of the disclosure and are provided solely for illustration of the aspects and not limitation thereof.

[0016] FIG. 1 illustrates an example wireless communications system, according to aspects of the disclosure.

[0017] FIGS. 2A and 2B illustrate example wireless network structures, according to aspects of the disclosure.

[0018] FIGS. 3A, 3B, and 3C are simplified block diagrams of several sample aspects of components that may be employed in a user equipment (UE), a base station, and a network entity, respectively, and configured to support communications as taught herein.

[0019] FIGS. 4A and 4B are diagrams illustrating example frame structures and channels within the frame structures, according to aspects of the disclosure.

[0020] FIG. 5 is a diagram of example positioning reference signal (PRS) resource sets having different time gaps, according to aspects of the disclosure.

[0021] FIGS. 6 to 8 illustrate various information elements (IEs) for reporting positioning measurements.

[0022] FIG. 9 is a diagram of several DL-PRS resources spanning a given time duration in milliseconds, according to aspects of the disclosure.

[0023] FIGS. 10 and 11 illustrate example methods of wireless positioning, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0024] Aspects of the disclosure are provided in the following description and related drawings directed to various examples provided for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.

[0025] The words “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.

[0026] Those of skill in the art will appreciate that the information and signals described below may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the description below may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.

[0027] Further, many aspects are described in terms of sequences of actions to be performed by, for example, elements of a computing device. It will be recognized that various actions described herein can be performed by specific circuits (e.g., application specific integrated circuits (ASICs)), by program instructions being executed by one or more processors, or by a combination of both. Additionally, the sequence(s) of actions described herein can be consid-

ered to be embodied entirely within any form of non-transitory computer-readable storage medium having stored therein a corresponding set of computer instructions that, upon execution, would cause or instruct an associated processor of a device to perform the functionality described herein. Thus, the various aspects of the disclosure may be embodied in a number of different forms, all of which have been contemplated to be within the scope of the claimed subject matter. In addition, for each of the aspects described herein, the corresponding form of any such aspects may be described herein as, for example, “logic configured to” perform the described action.

[0028] As used herein, the terms “user equipment” (UE) and “base station” are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, consumer asset locating device, wearable (e.g., smartwatch, glasses, augmented reality (AR)/virtual reality (VR) headset, etc.), vehicle (e.g., automobile, motorcycle, bicycle, etc.), Internet of Things (IoT) device, etc.) used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used herein, the term “UE” may be referred to interchangeably as an “access terminal” or “AT,” a “client device,” a “wireless device,” a “subscriber device,” a “subscriber terminal,” a “subscriber station,” a “user terminal” or “UT,” a “mobile device,” a “mobile terminal,” a “mobile station,” or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 specification, etc.) and so on.

[0029] A base station may operate according to one of several RATs in communication with UEs depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB, an evolved NodeB (eNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems a base station may provide purely edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, a forward traffic channel, etc.). As used herein the term traffic channel (TCH) can refer to either an uplink/reverse or downlink/forward traffic channel.

[0030] The term “base station” may refer to a single physical transmission-reception point (TRP) or to multiple physical TRPs that may or may not be co-located. For

example, where the term “base station” refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term “base station” refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term “base station” refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

[0031] In some implementations that support positioning of UEs, a base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit (e.g., when receiving and measuring signals from UEs).

[0032] An “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

[0033] FIG. 1 illustrates an example wireless communications system 100, according to aspects of the disclosure. The wireless communications system 100 (which may also be referred to as a wireless wide area network (WWAN)) may include various base stations 102 (labeled “BS”) and various UEs 104. The base stations 102 may include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base stations may include eNBs and/or ng-eNBs where the wireless communications system 100 corresponds to an LTE network, or gNBs where the wireless communications system 100 corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

[0034] The base stations 102 may collectively form a RAN and interface with a core network 170 (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links 122, and through the core network 170 to one or more

location servers **172** (e.g., a location management function (LMF) or a secure user plane location (SUPL) location platform (SLP)). The location server(s) **172** may be part of core network **170** or may be external to core network **170**. In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate with each other directly or indirectly (e.g., through the EPC/5GC) over backhaul links **134**, which may be wired or wireless.

[0035] The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each geographic coverage area **110**. A “cell” is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), an enhanced cell identifier (ECI), a virtual cell identifier (VCI), a cell global identifier (CGI), etc.) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term “cell” may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms “cell” and “TRP” may be used interchangeably. In some cases, the term “cell” may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.

[0036] While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102'** (labeled “SC” for “small cell”) may have a geographic coverage area **110'** that substantially overlaps with the geographic coverage area **110** of one or more macro cell base stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).

[0037] The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna

technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).

[0038] The wireless communications system **100** may further include a wireless local area network (WLAN) access point (AP) **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 GHz). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available.

[0039] The small cell base station **102'** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102'** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102'**, employing LTE/5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

[0040] The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and the UE **182** may utilize beamforming (transmit and/or receive) over a mmW communication link **184** to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations **102** may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

[0041] Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of

antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while cancelling to suppress radiation in undesired directions.

[0042] Transmit beams may be quasi-co-located, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically co-located. In NR, there are four types of quasi-co-location (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

[0043] In receive beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain in that direction of all other receive beams available to the receiver. This results in a stronger received signal strength (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

[0044] Transmit and receive beams may be spatially related. A spatial relation means that parameters for a second beam (e.g., a transmit or receive beam) for a second reference signal can be derived from information about a first beam (e.g., a receive beam or a transmit beam) for a first reference signal. For example, a UE may use a particular receive beam to receive a reference downlink reference signal (e.g., synchronization signal block (SSB)) from a base station. The UE can then form a transmit beam for sending an uplink reference signal (e.g., sounding reference signal (SRS)) to that base station based on the parameters of the receive beam.

[0045] Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the

downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a base station is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

[0046] In 5G, the frequency spectrum in which wireless nodes (e.g., base stations **102/180**, UEs **104/182**) operate is divided into multiple frequency ranges, FR1 (from 450 to 6000 MHz), FR2 (from 24250 to 52600 MHz), FR3 (above 52600 MHz), and FR4 (between FR1 and FR2). mmW frequency bands generally include the FR2, FR3, and FR4 frequency ranges. As such, the terms “mmW” and “FR2” or “FR3” or “FR4” may generally be used interchangeably.

[0047] In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104/182** and the cell in which the UE **104/182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that different UEs **104/182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104/182** at any time. This is done, for example, to balance the load on different carriers. Because a “serving cell” (whether a PCell or an SCell) corresponds to a carrier frequency/component carrier over which some base station is communicating, the term “cell,” “serving cell,” “component carrier,” “carrier frequency,” and the like can be used interchangeably.

[0048] For example, still referring to FIG. 1, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or “PCell”) and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers (“SCells”). The simultaneous transmission and/or reception of multiple carriers enables the UE **104/182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.

[0049] The wireless communications system **100** may further include a UE **164** that may communicate with a macro cell base station **102** over a communication link **120**

and/or the mmW base station **180** over a mmW communication link **184**. For example, the macro cell base station **102** may support a PCell and one or more SCells for the UE **164** and the mmW base station **180** may support one or more SCells for the UE **164**.

[0050] In the example of FIG. 1, one or more Earth orbiting satellite positioning system (SPS) space vehicles (SVs) **112** (e.g., satellites) may be used as an independent source of location information for any of the illustrated UEs (shown in FIG. 1 as a single UE **104** for simplicity). A UE **104** may include one or more dedicated SPS receivers specifically designed to receive SPS signals **124** for deriving geo location information from the SVs **112**. An SPS typically includes a system of transmitters (e.g., SVs **112**) positioned to enable receivers (e.g., UEs **104**) to determine their location on or above the Earth based, at least in part, on signals (e.g., SPS signals **124**) received from the transmitters. Such a transmitter typically transmits a signal marked with a repeating pseudo-random noise (PN) code of a set number of chips. While typically located in SVs **112**, transmitters may sometimes be located on ground-based control stations, base stations **102**, and/or other UEs **104**.

[0051] The use of SPS signals **124** can be augmented by various satellite-based augmentation systems (SBAS) that may be associated with or otherwise enabled for use with one or more global and/or regional navigation satellite systems. For example an SBAS may include an augmentation system(s) that provides integrity information, differential corrections, etc., such as the Wide Area Augmentation System (WAAS), the European Geostationary Navigation Overlay Service (EGNOS), the Multi-functional Satellite Augmentation System (MSAS), the Global Positioning System (GPS) Aided Geo Augmented Navigation or GPS and Geo Augmented Navigation system (GAGAN), and/or the like. Thus, as used herein, an SPS may include any combination of one or more global and/or regional navigation satellite systems and/or augmentation systems, and SPS signals **124** may include SPS, SPS-like, and/or other signals associated with such one or more SPS.

[0052] The wireless communications system **100** may further include one or more UEs, such as UE **190**, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as “sidelinks”). In the example of FIG. 1, UE **190** has a D2D P2P link **192** with one of the UEs **104** connected to one of the base stations **102** (e.g., through which UE **190** may indirectly obtain cellular connectivity) and a D2D P2P link **194** with WLAN STA **152** connected to the WLAN AP **150** (through which UE **190** may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links **192** and **194** may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), WiFi Direct (WiFi-D), Bluetooth®, and so on.

[0053] FIG. 2A illustrates an example wireless network structure **200**. For example, a 5GC **210** (also referred to as a Next Generation Core (NGC)) can be viewed functionally as control plane (C-plane) functions **214** (e.g., UE registration, authentication, network access, gateway selection, etc.) and user plane (U-plane) functions **212**, (e.g., UE gateway function, access to data networks, IP routing, etc.) which operate cooperatively to form the core network. User plane interface (NG-U) **213** and control plane interface (NG-C) **215** connect the gNB **222** to the 5GC **210** and specifically to the user plane functions **212** and control plane functions **214**,

respectively. In an additional configuration, an ng-eNB **224** may also be connected to the 5GC **210** via NG-C **215** to the control plane functions **214** and NG-U **213** to user plane functions **212**. Further, ng-eNB **224** may directly communicate with gNB **222** via a backhaul connection **223**. In some configurations, a Next Generation RAN (NG-RAN) **220** may have one or more gNBs **222**, while other configurations include one or more of both ng-eNBs **224** and gNBs **222**. Either (or both) gNB **222** or ng-eNB **224** may communicate with one or more UEs **204** (e.g., any of the UEs described herein).

[0054] Another optional aspect may include a location server **230**, which may be in communication with the 5GC **210** to provide location assistance for UE(s) **204**. The location server **230** can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The location server **230** can be configured to support one or more location services for UEs **204** that can connect to the location server **230** via the core network, 5GC **210**, and/or via the Internet (not illustrated). Further, the location server **230** may be integrated into a component of the core network, or alternatively may be external to the core network (e.g., a third party server, such as an original equipment manufacturer (OEM) server or service server).

[0055] FIG. 2B illustrates another example wireless network structure **250**. A 5GC **260** (which may correspond to 5GC **210** in FIG. 2A) can be viewed functionally as control plane functions, provided by an access and mobility management function (AMF) **264**, and user plane functions, provided by a user plane function (UPF) **262**, which operate cooperatively to form the core network (i.e., 5GC **260**). The functions of the AMF **264** include registration management, connection management, reachability management, mobility management, lawful interception, transport for session management (SM) messages between one or more UEs **204** (e.g., any of the UEs described herein) and a session management function (SMF) **266**, transparent proxy services for routing SM messages, access authentication and access authorization, transport for short message service (SMS) messages between the UE **204** and the short message service function (SMSF) (not shown), and security anchor functionality (SEAF). The AMF **264** also interacts with an authentication server function (AUSF) (not shown) and the UE **204**, and receives the intermediate key that was established as a result of the UE **204** authentication process. In the case of authentication based on a UMTS (universal mobile telecommunications system) subscriber identity module (USIM), the AMF **264** retrieves the security material from the AUSF. The functions of the AMF **264** also include security context management (SCM). The SCM receives a key from the SEAF that it uses to derive access-network specific keys. The functionality of the AMF **264** also includes location services management for regulatory services, transport for location services messages between the UE **204** and a location management function (LMF) **270** (which acts as a location server **230**), transport for location services messages between the NG-RAN **220** and the LMF **270**, evolved packet system (EPS) bearer identifier allocation for interworking with the EPS, and UE **204** mobility

event notification. In addition, the AMF 264 also supports functionalities for non-3GPP (Third Generation Partnership Project) access networks.

[0056] Functions of the UPF 262 include acting as an anchor point for intra-/inter-RAT mobility (when applicable), acting as an external protocol data unit (PDU) session point of interconnect to a data network (not shown), providing packet routing and forwarding, packet inspection, user plane policy rule enforcement (e.g., gating, redirection, traffic steering), lawful interception (user plane collection), traffic usage reporting, quality of service (QoS) handling for the user plane (e.g., uplink/downlink rate enforcement, reflective QoS marking in the downlink), uplink traffic verification (service data flow (SDF) to QoS flow mapping), transport level packet marking in the uplink and downlink, downlink packet buffering and downlink data notification triggering, and sending and forwarding of one or more “end markers” to the source RAN node. The UPF 262 may also support transfer of location services messages over a user plane between the UE 204 and a location server, such as an SLP 272.

[0057] The functions of the SMF 266 include session management, UE Internet protocol (IP) address allocation and management, selection and control of user plane functions, configuration of traffic steering at the UPF 262 to route traffic to the proper destination, control of part of policy enforcement and QoS, and downlink data notification. The interface over which the SMF 266 communicates with the AMF 264 is referred to as the N11 interface.

[0058] Another optional aspect may include an LMF 270, which may be in communication with the 5GC 260 to provide location assistance for UEs 204. The LMF 270 can be implemented as a plurality of separate servers (e.g., physically separate servers, different software modules on a single server, different software modules spread across multiple physical servers, etc.), or alternately may each correspond to a single server. The LMF 270 can be configured to support one or more location services for UEs 204 that can connect to the LMF 270 via the core network, 5GC 260, and/or via the Internet (not illustrated). The SLP 272 may support similar functions to the LMF 270, but whereas the LMF 270 may communicate with the AMF 264, NG-RAN 220, and UEs 204 over a control plane (e.g., using interfaces and protocols intended to convey signaling messages and not voice or data), the SLP 272 may communicate with UEs 204 and external clients (not shown in FIG. 2B) over a user plane (e.g., using protocols intended to carry voice and/or data like the transmission control protocol (TCP) and/or IP).

[0059] User plane interface 263 and control plane interface 265 connect the 5GC 260, and specifically the UPF 262 and AMF 264, respectively, to one or more gNBs 222 and/or ng-eNBs 224 in the NG-RAN 220. The interface between gNB(s) 222 and/or ng-eNB(s) 224 and the AMF 264 is referred to as the “N2” interface, and the interface between gNB(s) 222 and/or ng-eNB(s) 224 and the UPF 262 is referred to as the “N3” interface.

[0060] The gNB(s) 222 and/or ng-eNB(s) 224 of the NG-RAN 220 may communicate directly with each other via backhaul connections 223, referred to as the “Xn-C” interface. One or more of gNBs 222 and/or ng-eNBs 224 may communicate with one or more UEs 204 over a wireless interface, referred to as the “Uu” interface.

[0061] The functionality of a gNB 222 is divided between a gNB central unit (gNB-CU) 226 and one or more gNB

distributed units (gNB-DUs) 228. The interface 232 between the gNB-CU 226 and the one or more gNB-DUs 228 is referred to as the “F1” interface. A gNB-CU 226 is a logical node that includes the base station functions of transferring user data, mobility control, radio access network sharing, positioning, session management, and the like, except for those functions allocated exclusively to the gNB-DU(s) 228. More specifically, the gNB-CU 226 hosts the radio resource control (RRC), service data adaptation protocol (SDAP), and packet data convergence protocol (PDCP) protocols of the gNB 222. A gNB-DU 228 is a logical node that hosts the radio link control (RLC), medium access control (MAC), and physical (PHY) layers of the gNB 222. Its operation is controlled by the gNB-CU 226. One gNB-DU 228 can support one or more cells, and one cell is supported by only one gNB-DU 228. Thus, a UE 204 communicates with the gNB-CU 226 via the RRC, SDAP, and PDCP layers and with a gNB-DU 228 via the RLC, MAC, and PHY layers.

[0062] FIGS. 3A, 3B, and 3C illustrate several example components (represented by corresponding blocks) that may be incorporated into a UE 302 (which may correspond to any of the UEs described herein), a base station 304 (which may correspond to any of the base stations described herein), and a network entity 306 (which may correspond to or embody any of the network functions described herein, including the location server 230 and the LMF 270, or alternatively may be independent from the NG-RAN 220 and/or 5GC 210/260 infrastructure depicted in FIGS. 2A and 2B, such as a private network) to support the file transmission operations as taught herein. It will be appreciated that these components may be implemented in different types of apparatuses in different implementations (e.g., in an ASIC, in a system-on-chip (SoC), etc.). The illustrated components may also be incorporated into other apparatuses in a communication system. For example, other apparatuses in a system may include components similar to those described to provide similar functionality. Also, a given apparatus may contain one or more of the components. For example, an apparatus may include multiple transceiver components that enable the apparatus to operate on multiple carriers and/or communicate via different technologies.

[0063] The UE 302 and the base station 304 each include one or more wireless wide area network (WWAN) transceivers 310 and 350, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) via one or more wireless communication networks (not shown), such as an NR network, an LTE network, a GSM network, and/or the like. The WWAN transceivers 310 and 350 may each be connected to one or more antennas 316 and 356, respectively, for communicating with other network nodes, such as other UEs, access points, base stations (e.g., eNBs, gNBs), etc., via at least one designated RAT (e.g., NR, LTE, GSM, etc.) over a wireless communication medium of interest (e.g., some set of time/frequency resources in a particular frequency spectrum). The WWAN transceivers 310 and 350 may be variously configured for transmitting and encoding signals 318 and 358 (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals 318 and 358 (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the WWAN transceivers 310 and 350 include one or more transmitters 314 and 354,

respectively, for transmitting and encoding signals **318** and **358**, respectively, and one or more receivers **312** and **352**, respectively, for receiving and decoding signals **318** and **358**, respectively.

[0064] The UE **302** and the base station **304** each also include, at least in some cases, one or more short-range wireless transceivers **320** and **360**, respectively. The short-range wireless transceivers **320** and **360** may be connected to one or more antennas **326** and **366**, respectively, and provide means for communicating (e.g., means for transmitting, means for receiving, means for measuring, means for tuning, means for refraining from transmitting, etc.) with other network nodes, such as other UEs, access points, base stations, etc., via at least one designated RAT (e.g., WiFi, LTE-D, Bluetooth®, Zigbee®, Z-Wave®, PC5, dedicated short-range communications (DSRC), wireless access for vehicular environments (WAVE), near-field communication (NFC), etc.) over a wireless communication medium of interest. The short-range wireless transceivers **320** and **360** may be variously configured for transmitting and encoding signals **328** and **368** (e.g., messages, indications, information, and so on), respectively, and, conversely, for receiving and decoding signals **328** and **368** (e.g., messages, indications, information, pilots, and so on), respectively, in accordance with the designated RAT. Specifically, the short-range wireless transceivers **320** and **360** include one or more transmitters **324** and **364**, respectively, for transmitting and encoding signals **328** and **368**, respectively, and one or more receivers **322** and **362**, respectively, for receiving and decoding signals **328** and **368**, respectively. As specific examples, the short-range wireless transceivers **320** and **360** may be WiFi transceivers, Bluetooth® transceivers, Zigbee® and/or Z-Wave® transceivers, NFC transceivers, or vehicle-to-vehicle (V2V) and/or vehicle-to-everything (V2X) transceivers.

[0065] The UE **302** and the base station **304** also include, at least in some cases, satellite positioning systems (SPS) receivers **330** and **370**. The SPS receivers **330** and **370** may be connected to one or more antennas **336** and **376**, respectively, and may provide means for receiving and/or measuring SPS signals **338** and **378**, respectively, such as global positioning system (GPS) signals, global navigation satellite system (GLONASS) signals, Galileo signals, Beidou signals, Indian Regional Navigation Satellite System (NAVIC), Quasi-Zenith Satellite System (QZSS), etc. The SPS receivers **330** and **370** may comprise any suitable hardware and/or software for receiving and processing SPS signals **338** and **378**, respectively. The SPS receivers **330** and **370** request information and operations as appropriate from the other systems, and performs calculations necessary to determine positions of the UE **302** and the base station **304** using measurements obtained by any suitable SPS algorithm.

[0066] The base station **304** and the network entity **306** each include one or more network transceivers **380** and **390**, respectively, providing means for communicating (e.g., means for transmitting, means for receiving, etc.) with other network entities (e.g., other base stations **304**, other network entities **306**). For example, the base station **304** may employ the one or more network transceivers **380** to communicate with other base stations **304** or network entities **306** over one or more wired or wireless backhaul links. As another example, the network entity **306** may employ the one or more network transceivers **390** to communicate with one or more base station **304** over one or more wired or wireless

backhaul links, or with other network entities **306** over one or more wired or wireless core network interfaces.

[0067] A transceiver may be configured to communicate over a wired or wireless link. A transceiver (whether a wired transceiver or a wireless transceiver) includes transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) and receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**). A transceiver may be an integrated device (e.g., embodying transmitter circuitry and receiver circuitry in a single device) in some implementations, may comprise separate transmitter circuitry and separate receiver circuitry in some implementations, or may be embodied in other ways in other implementations. The transmitter circuitry and receiver circuitry of a wired transceiver (e.g., network transceivers **380** and **390** in some implementations) may be coupled to one or more wired network interface ports. Wireless transmitter circuitry (e.g., transmitters **314**, **324**, **354**, **364**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform transmit “beamforming,” as described herein. Similarly, wireless receiver circuitry (e.g., receivers **312**, **322**, **352**, **362**) may include or be coupled to a plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such as an antenna array, that permits the respective apparatus (e.g., UE **302**, base station **304**) to perform receive beamforming, as described herein. In an aspect, the transmitter circuitry and receiver circuitry may share the same plurality of antennas (e.g., antennas **316**, **326**, **356**, **366**), such that the respective apparatus can only receive or transmit at a given time, not both at the same time. A wireless transceiver (e.g., WWAN transceivers **310** and **350**, short-range wireless transceivers **320** and **360**) may also include a network listen module (NLM) or the like for performing various measurements.

[0068] As used herein, the various wireless transceivers (e.g., transceivers **310**, **320**, **350**, and **360**, and network transceivers **380** and **390** in some implementations) and wired transceivers (e.g., network transceivers **380** and **390** in some implementations) may generally be characterized as “a transceiver,” “at least one transceiver,” or “one or more transceivers.” As such, whether a particular transceiver is a wired or wireless transceiver may be inferred from the type of communication performed. For example, backhaul communication between network devices or servers will generally relate to signaling via a wired transceiver, whereas wireless communication between a UE (e.g., UE **302**) and a base station (e.g., base station **304**) will generally relate to signaling via a wireless transceiver.

[0069] The UE **302**, the base station **304**, and the network entity **306** also include other components that may be used in conjunction with the operations as disclosed herein. The UE **302**, the base station **304**, and the network entity **306** include one or more processors **332**, **384**, and **394**, respectively, for providing functionality relating to, for example, wireless communication, and for providing other processing functionality. The processors **332**, **384**, and **394** may therefore provide means for processing, such as means for determining, means for calculating, means for receiving, means for transmitting, means for indicating, etc. In an aspect, the processors **332**, **384**, and **394** may include, for example, one or more general purpose processors, multi-core processors, central processing units (CPUs), ASICs, digital signal processors (DSPs), field programmable gate

arrays (FPGAs), other programmable logic devices or processing circuitry, or various combinations thereof.

[0070] The UE 302, the base station 304, and the network entity 306 include memory circuitry implementing memories 340, 386, and 396 (e.g., each including a memory device), respectively, for maintaining information (e.g., information indicative of reserved resources, thresholds, parameters, and so on). The memories 340, 386, and 396 may therefore provide means for storing, means for retrieving, means for maintaining, etc. In some cases, the UE 302, the base station 304, and the network entity 306 may include positioning component 342, 388, and 398, respectively. The positioning component 342, 388, and 398 may be hardware circuits that are part of or coupled to the processors 332, 384, and 394, respectively, that, when executed, cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. In other aspects, the positioning component 342, 388, and 398 may be external to the processors 332, 384, and 394 (e.g., part of a modem processing system, integrated with another processing system, etc.). Alternatively, the positioning component 342, 388, and 398 may be memory modules stored in the memories 340, 386, and 396, respectively, that, when executed by the processors 332, 384, and 394 (or a modem processing system, another processing system, etc.), cause the UE 302, the base station 304, and the network entity 306 to perform the functionality described herein. FIG. 3A illustrates possible locations of the positioning component 342, which may be, for example, part of the one or more WWAN transceivers 310, the memory 340, the one or more processors 332, or any combination thereof, or may be a standalone component. FIG. 3B illustrates possible locations of the positioning component 388, which may be, for example, part of the one or more WWAN transceivers 350, the memory 386, the one or more processors 384, or any combination thereof, or may be a standalone component. FIG. 3C illustrates possible locations of the positioning component 398, which may be, for example, part of the one or more network transceivers 390, the memory 396, the one or more processors 394, or any combination thereof, or may be a standalone component.

[0071] The UE 302 may include one or more sensors 344 coupled to the one or more processors 332 to provide means for sensing or detecting movement and/or orientation information that is independent of motion data derived from signals received by the one or more WWAN transceivers 310, the one or more short-range wireless transceivers 320, and/or the SPS receiver 330. By way of example, the sensor(s) 344 may include an accelerometer (e.g., a micro-electrical mechanical systems (MEMS) device), a gyroscope, a geomagnetic sensor (e.g., a compass), an altimeter (e.g., a barometric pressure altimeter), and/or any other type of movement detection sensor. Moreover, the sensor(s) 344 may include a plurality of different types of devices and combine their outputs in order to provide motion information. For example, the sensor(s) 344 may use a combination of a multi-axis accelerometer and orientation sensors to provide the ability to compute positions in two-dimensional (2D) and/or three-dimensional (3D) coordinate systems.

[0072] In addition, the UE 302 includes a user interface 346 providing means for providing indications (e.g., audible and/or visual indications) to a user and/or for receiving user input (e.g., upon user actuation of a sensing device such as a keypad, a touch screen, a microphone, and so on). Although

not shown, the base station 304 and the network entity 306 may also include user interfaces.

[0073] Referring to the one or more processors 384 in more detail, in the downlink, IP packets from the network entity 306 may be provided to the processor 384. The one or more processors 384 may implement functionality for an RRC layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The one or more processors 384 may provide RRC layer functionality associated with broadcasting of system information (e.g., master information block (MIB), system information blocks (SIBs)), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter-RAT mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer PDUs, error correction through automatic repeat request (ARQ), concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, scheduling information reporting, error correction, priority handling, and logical channel prioritization.

[0074] The transmitter 354 and the receiver 352 may implement Layer-1 (L1) functionality associated with various signal processing functions. Layer-1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The transmitter 354 handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an orthogonal frequency division multiplexing (OFDM) subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an inverse fast Fourier transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM symbol stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE 302. Each spatial stream may then be provided to one or more different antennas 356. The transmitter 354 may modulate an RF carrier with a respective spatial stream for transmission.

[0075] At the UE 302, the receiver 312 receives a signal through its respective antenna(s) 316. The receiver 312 recovers information modulated onto an RF carrier and provides the information to the one or more processors 332. The transmitter 314 and the receiver 312 implement Layer-1 functionality associated with various signal processing functions. The receiver 312 may perform spatial processing on

the information to recover any spatial streams destined for the UE 302. If multiple spatial streams are destined for the UE 302, they may be combined by the receiver 312 into a single OFDM symbol stream. The receiver 312 then converts the OFDM symbol stream from the time-domain to the frequency domain using a fast Fourier transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station 304. These soft decisions may be based on channel estimates computed by a channel estimator. The soft decisions are then decoded and de-interleaved to recover the data and control signals that were originally transmitted by the base station 304 on the physical channel. The data and control signals are then provided to the one or more processors 332, which implements Layer-3 (L3) and Layer-2 (L2) functionality.

[0076] In the uplink, the one or more processors 332 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the core network. The one or more processors 332 are also responsible for error detection.

[0077] Similar to the functionality described in connection with the downlink transmission by the base station 304, the one or more processors 332 provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through hybrid automatic repeat request (HARQ), priority handling, and logical channel prioritization.

[0078] Channel estimates derived by the channel estimator from a reference signal or feedback transmitted by the base station 304 may be used by the transmitter 314 to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the transmitter 314 may be provided to different antenna(s) 316. The transmitter 314 may modulate an RF carrier with a respective spatial stream for transmission.

[0079] The uplink transmission is processed at the base station 304 in a manner similar to that described in connection with the receiver function at the UE 302. The receiver 352 receives a signal through its respective antenna(s) 356. The receiver 352 recovers information modulated onto an RF carrier and provides the information to the one or more processors 384.

[0080] In the uplink, the one or more processors 384 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE 302. IP packets from the one or more processors 384 may be provided to the core network. The one or more processors 384 are also responsible for error detection.

[0081] For convenience, the UE 302, the base station 304, and/or the network entity 306 are shown in FIGS. 3A, 3B, and 3C as including various components that may be configured according to the various examples described herein. It will be appreciated, however, that the illustrated components may have different functionality in different designs. In particular, various components in FIGS. 3A to 3C are optional in alternative configurations and the various aspects include configurations that may vary due to design choice, costs, use of the device, or other considerations. For example, in case of FIG. 3A, a particular implementation of UE 302 may omit the WWAN transceiver(s) 310 (e.g., a wearable device or tablet computer or PC or laptop may have Wi-Fi and/or Bluetooth capability without cellular capability), or may omit the short-range wireless transceiver(s) 320 (e.g., cellular-only, etc.), or may omit the SPS receiver 330, or may omit the sensor(s) 344, and so on. In another example, in case of FIG. 3B, a particular implementation of the base station 304 may omit the WWAN transceiver(s) 350 (e.g., a Wi-Fi “hotspot” access point without cellular capability), or may omit the short-range wireless transceiver(s) 360 (e.g., cellular-only, etc.), or may omit the SPS receiver 370, and so on. For brevity, illustration of the various alternative configurations is not provided herein, but would be readily understandable to one skilled in the art.

[0082] The various components of the UE 302, the base station 304, and the network entity 306 may be communicatively coupled to each other over data buses 334, 382, and 392, respectively. In an aspect, the data buses 334, 382, and 392 may form, or be part of, a communication interface of the UE 302, the base station 304, and the network entity 306, respectively. For example, where different logical entities are embodied in the same device (e.g., gNB and location server functionality incorporated into the same base station 304), the data buses 334, 382, and 392 may provide communication between them.

[0083] The components of FIGS. 3A, 3B, and 3C may be implemented in various ways. In some implementations, the components of FIGS. 3A, 3B, and 3C may be implemented in one or more circuits such as, for example, one or more processors and/or one or more ASICs (which may include one or more processors). Here, each circuit may use and/or incorporate at least one memory component for storing information or executable code used by the circuit to provide this functionality. For example, some or all of the functionality represented by blocks 310 to 346 may be implemented by processor and memory component(s) of the UE 302 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Similarly, some or all of the functionality represented by blocks 350 to 388 may be implemented by processor and memory component(s) of the base station 304 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). Also, some or all of the functionality represented by blocks 390 to 398 may be implemented by processor and memory component(s) of the network entity 306 (e.g., by execution of appropriate code and/or by appropriate configuration of processor components). For simplicity, various operations, acts, and/or functions are described herein as being performed “by a UE,” “by a base station,” “by a network entity,” etc. However, as will be appreciated, such operations, acts, and/or functions may actually be performed by specific components or combinations of components of the UE 302, base station 304, network entity 306, etc., such

as the processors 332, 384, 394, the transceivers 310, 320, 350, and 360, the memories 340, 386, and 396, the positioning component 342, 388, and 398, etc.

[0084] In some designs, the network entity 306 may be implemented as a core network component. In other designs, the network entity 306 may be distinct from a network operator or operation of the cellular network infrastructure (e.g., NG RAN 220 and/or 5GC 210/260). For example, the network entity 306 may be a component of a private network that may be configured to communicate with the UE 302 via the base station 304 or independently from the base station 304 (e.g., over a non-cellular communication link, such as WiFi).

[0085] NR supports a number of cellular network-based positioning technologies, including downlink-based, uplink-based, and downlink-and-uplink-based positioning methods. Downlink-based positioning methods include observed time difference of arrival (OTDOA) in LTE, downlink time difference of arrival (DL-TDOA) in NR, and downlink angle-of-departure (DL-AoD) in NR. In an OTDOA or DL-TDOA positioning procedure, a UE measures the differences between the times of arrival (ToAs) of reference signals (e.g., positioning reference signals (PRS)) received from pairs of base stations, referred to as reference signal time difference (RSTD) or time difference of arrival (TDOA) measurements, and reports them to a positioning entity. More specifically, the UE receives the identifiers (IDs) of a reference base station (e.g., a serving base station) and multiple non-reference base stations in assistance data. The UE then measures the RSTD between the reference base station and each of the non-reference base stations. Based on the known locations of the involved base stations and the RSTD measurements, the positioning entity can estimate the UE's location.

[0086] For DL-AoD positioning, the positioning entity uses a beam report from the UE of received signal strength measurements of multiple downlink transmit beams to determine the angle(s) between the UE and the transmitting base station(s). The positioning entity can then estimate the location of the UE based on the determined angle(s) and the known location(s) of the transmitting base station(s).

[0087] Uplink-based positioning methods include uplink time difference of arrival (UL-TDOA) and uplink angle-of-arrival (UL-AoA). UL-TDOA is similar to DL-TDOA, but is based on uplink reference signals (e.g., sounding reference signals (SRS)) transmitted by the UE. For UL-AoA positioning, one or more base stations measure the received signal strength of one or more uplink reference signals (e.g., SRS) received from a UE on one or more uplink receive beams. The positioning entity uses the signal strength measurements and the angle(s) of the receive beam(s) to determine the angle(s) between the UE and the base station(s). Based on the determined angle(s) and the known location(s) of the base station(s), the positioning entity can then estimate the location of the UE.

[0088] Downlink-and-uplink-based positioning methods include enhanced cell-ID (E-CID) positioning and multi-round-trip-time (RTT) positioning (also referred to as "multi-cell RTT"). In an RTT procedure, an initiator (a base station or a UE) transmits an RTT measurement signal (e.g., a PRS or SRS) to a responder (a UE or base station), which transmits an RTT response signal (e.g., an SRS or PRS) back to the initiator. The RTT response signal includes the difference between the ToA of the RTT measurement signal

and the transmission time of the RTT response signal, referred to as the reception-to-transmission (Rx-Tx) time difference. The initiator calculates the difference between the transmission time of the RTT measurement signal and the ToA of the RTT response signal, referred to as the transmission-to-reception (Tx-Rx) time difference. The propagation time (also referred to as the "time of flight") between the initiator and the responder can be calculated from the Tx-Rx and Rx-Tx time differences. Based on the propagation time and the known speed of light, the distance between the initiator and the responder can be determined. For multi-RTT positioning, a UE performs an RTT procedure with multiple base stations to enable its location to be determined (e.g., using multilateration) based on the known locations of the base stations. RTT and multi-RTT methods can be combined with other positioning techniques, such as UL-AoA and DL-AoD, to improve location accuracy.

[0089] The E-CID positioning method is based on radio resource management (RRM) measurements. In E-CID, the UE reports the serving cell ID, the timing advance (TA), and the identifiers, estimated timing, and signal strength of detected neighbor base stations. The location of the UE is then estimated based on this information and the known locations of the base station(s).

[0090] To assist positioning operations, a location server (e.g., location server 230, LMF 270, SLP 272) may provide assistance data to the UE. For example, the assistance data may include identifiers of the base stations (or the cells/TRPs of the base stations) from which to measure reference signals, the reference signal configuration parameters (e.g., the number of consecutive positioning subframes, periodicity of positioning subframes, muting sequence, frequency hopping sequence, reference signal identifier, reference signal bandwidth, etc.), and/or other parameters applicable to the particular positioning method. Alternatively, the assistance data may originate directly from the base stations themselves (e.g., in periodically broadcasted overhead messages, etc.). In some cases, the UE may be able to detect neighbor network nodes itself without the use of assistance data.

[0091] In the case of an OTDOA or DL-TDOA positioning procedure, the assistance data may further include an expected RSTD value and an associated uncertainty, or search window, around the expected RSTD. In some cases, the value range of the expected RSTD may be ± 500 microseconds (μ s). In some cases, when any of the resources used for the positioning measurement are in FR1, the value range for the uncertainty of the expected RSTD may be ± 32 μ s. In other cases, when all of the resources used for the positioning measurement(s) are in FR2, the value range for the uncertainty of the expected RSTD may be ± 8 μ s.

[0092] A location estimate may be referred to by other names, such as a position estimate, location, position, position fix, fix, or the like. A location estimate may be geodetic and comprise coordinates (e.g., latitude, longitude, and possibly altitude) or may be civic and comprise a street address, postal address, or some other verbal description of a location. A location estimate may further be defined relative to some other known location or defined in absolute terms (e.g., using latitude, longitude, and possibly altitude). A location estimate may include an expected error or uncertainty (e.g., by including an area or volume within which the location is expected to be included with some specified or default level of confidence).

[0093] Various frame structures may be used to support downlink and uplink transmissions between network nodes (e.g., base stations and UEs). FIG. 4A is a diagram 400 illustrating an example of a downlink frame structure, according to aspects of the disclosure. FIG. 4B is a diagram 430 illustrating an example of channels within the downlink frame structure, according to aspects of the disclosure. Other wireless communications technologies may have different frame structures and/or different channels.

[0094] LTE, and in some cases NR, utilizes OFDM on the downlink and single-carrier frequency division multiplexing (SC-FDM) on the uplink. Unlike LTE, however, NR has an option to use OFDM on the uplink as well. OFDM and SC-FDM partition the system bandwidth into multiple (K) orthogonal subcarriers, which are also commonly referred to as tones, bins, etc. Each subcarrier may be modulated with data. In general, modulation symbols are sent in the frequency domain with OFDM and in the time domain with SC-FDM. The spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system bandwidth. For example, the spacing of the subcarriers may be 15 kilohertz (kHz) and the minimum resource allocation (resource block) may be 12 subcarriers (or 180 kHz). Consequently, the nominal FFT size may be equal to 128, 256, 512, 1024, or 2048 for system bandwidth of 1.25, 2.5, 5, 10, or 20 megahertz (MHz), respectively. The system bandwidth may also be partitioned into subbands. For example, a subband may cover 1.08 MHz (i.e., 6 resource blocks), and there may be 1, 2, 4, 8, or 16 subbands for system bandwidth of 1.25, 2.5, 5, 10, or 20 MHz, respectively.

[0095] LTE supports a single numerology (subcarrier spacing (SCS), symbol length, etc.). In contrast, NR may support multiple numerologies (μ), for example, subcarrier spacings of 15 kHz ($\mu=0$), 30 kHz ($\mu=1$), 60 kHz ($\mu=2$), 120 kHz ($\mu=3$), and 240 kHz ($\mu=4$) or greater may be available. In each subcarrier spacing, there are 14 symbols per slot. For 15 kHz SCS ($p=0$), there is one slot per subframe, 10 slots per frame, the slot duration is 1 millisecond (ms), the symbol duration is 66.7 microseconds (μ s), and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 50. For 30 kHz SCS ($p=1$), there are two slots per subframe, 20 slots per frame, the slot duration is 0.5 ms, the symbol duration is 33.3 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 100. For 60 kHz SCS ($\mu=2$), there are four slots per subframe, 40 slots per frame, the slot duration is 0.25 ms, the symbol duration is 16.7 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 200. For 120 kHz SCS ($\mu=3$), there are eight slots per subframe, 80 slots per frame, the slot duration is 0.125 ms, the symbol duration is 8.33 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 400. For 240 kHz SCS ($\mu=4$), there are 16 slots per subframe, 160 slots per frame, the slot duration is 0.0625 ms, the symbol duration is 4.17 μ s, and the maximum nominal system bandwidth (in MHz) with a 4K FFT size is 800.

[0096] In the example of FIGS. 4A and 4B, a numerology of 15 kHz is used. Thus, in the time domain, a 10 ms frame is divided into 10 equally sized subframes of 1 ms each, and each subframe includes one time slot. In FIGS. 4A and 4B, time is represented horizontally (on the X axis) with time increasing from left to right, while frequency is represented

vertically (on the Y axis) with frequency increasing (or decreasing) from bottom to top.

[0097] A resource grid may be used to represent time slots, each time slot including one or more time-concurrent resource blocks (RBs) (also referred to as physical RBs (PRBs)) in the frequency domain. The resource grid is further divided into multiple resource elements (REs). An RE may correspond to one symbol length in the time domain and one subcarrier in the frequency domain. In the nomenclature of FIGS. 4A and 4B, for a normal cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and seven consecutive symbols in the time domain, for a total of 84 REs. For an extended cyclic prefix, an RB may contain 12 consecutive subcarriers in the frequency domain and six consecutive symbols in the time domain, for a total of 72 REs. The number of bits carried by each RE depends on the modulation scheme.

[0098] Some of the REs carry downlink reference (pilot) signals (DL-RS). The DL-RS may include positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signals (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), demodulation reference signals (DMRS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), etc. FIG. 4A illustrates example locations of REs carrying PRS (labeled “R”).

[0099] A collection of resource elements (REs) that are used for transmission of PRS is referred to as a “PRS resource.” The collection of resource elements can span multiple PRBs in the frequency domain and ‘N’ (such as 1 or more) consecutive symbol(s) within a slot in the time domain. In a given OFDM symbol in the time domain, a PRS resource occupies consecutive PRBs in the frequency domain.

[0100] The transmission of a PRS resource within a given PRB has a particular comb size (also referred to as the “comb density”). A comb size ‘N’ represents the subcarrier spacing (or frequency/tone spacing) within each symbol of a PRS resource configuration.

[0101] Specifically, for a comb size ‘N,’ PRS are transmitted in every Nth subcarrier of a symbol of a PRB. For example, for comb-4, for each symbol of the PRS resource configuration, REs corresponding to every fourth subcarrier (such as subcarriers 0, 4, 8) are used to transmit PRS of the PRS resource. Currently, comb sizes of comb-2, comb-4, comb-6, and comb-12 are supported for DL-PRS. FIG. 4A illustrates an example PRS resource configuration for comb-6 (which spans six symbols). That is, the locations of the shaded REs (labeled “R”) indicate a comb-6 PRS resource configuration.

[0102] Currently, a DL-PRS resource may span 2, 4, 6, or 12 consecutive symbols within a slot with a fully frequency-domain staggered pattern. A DL-PRS resource can be configured in any higher layer configured downlink or flexible (FL) symbol of a slot. There may be a constant energy per resource element (EPRE) for all REs of a given DL-PRS resource.

[0103] The following are the frequency offsets from symbol to symbol for comb sizes 2, 4, 6, and 12 over 2, 4, 6, and 12 symbols. 2-symbol comb-2: {0, 1}; 4-symbol comb-2: {0, 1, 0, 1}; 6-symbol comb-2: {0, 1, 0, 1, 0, 1}; 12-symbol comb-2: {0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1}; 4-symbol comb-4: {0, 2, 1, 3}; 12-symbol comb-4: {0, 2, 1, 3, 0, 2, 1, 3, 0, 2, 1, 3};

1, 3}; 6-symbol comb-6: {0, 3, 1, 4, 2, 5}; 12-symbol comb-6: {0, 3, 1, 4, 2, 5, 0, 3, 1, 4, 2, 5}; and 12-symbol comb-12: {0, 6, 3, 9, 1, 7, 4, 10, 2, 8, 5, 11}.

[0104] A “PRS resource set” is a set of PRS resources used for the transmission of PRS signals, where each PRS resource has a PRS resource ID. In addition, the PRS resources in a PRS resource set are associated with the same TRP. A PRS resource set is identified by a PRS resource set ID and is associated with a particular TRP (identified by a TRP ID). In addition, the PRS resources in a PRS resource set have the same periodicity, a common muting pattern configuration, and the same repetition factor (such as “PRS-ResourceRepetitionFactor”) across slots. The periodicity is the time from the first repetition of the first PRS resource of a first PRS instance to the same first repetition of the same first PRS resource of the next PRS instance. The periodicity may have a length selected from $2^{\mu} \times \{4, 5, 8, 10, 16, 20, 32, 40, 64, 80, 160, 320, 640, 1280, 2560, 5120, 10240\}$ slots, with $\mu=0, 1, 2, 3$. The repetition factor may have a length selected from $\{1, 2, 4, 6, 8, 16, 32\}$ slots.

[0105] A PRS resource ID in a PRS resource set is associated with a single beam (or beam ID) transmitted from a single TRP (where a TRP may transmit one or more beams). That is, each PRS resource of a PRS resource set may be transmitted on a different beam, and as such, a “PRS resource,” or simply “resource,” also can be referred to as a “beam.” Note that this does not have any implications on whether the TRPs and the beams on which PRS are transmitted are known to the UE.

[0106] A “PRS instance” or “PRS occasion” is one instance of a periodically repeated time window (such as a group of one or more consecutive slots) where PRS are expected to be transmitted. A PRS occasion also may be referred to as a “PRS positioning occasion,” a “PRS positioning instance, a “positioning occasion,” “a positioning instance,” a “positioning repetition,” or simply an “occasion,” an “instance,” or a “repetition.”

[0107] A “positioning frequency layer” (also referred to simply as a “frequency layer”) is a collection of one or more PRS resource sets across one or more TRPs that have the same values for certain parameters. Specifically, the collection of PRS resource sets has the same subcarrier spacing and cyclic prefix (CP) type (meaning all numerologies supported for the PDSCH are also supported for PRS), the same Point A, the same value of the downlink PRS bandwidth, the same start PRB (and center frequency), and the same comb-size. The Point A parameter takes the value of the parameter “ARFCN-ValueNR” (where “ARFCN” stands for “absolute radio-frequency channel number”) and is an identifier/code that specifies a pair of physical radio channel used for transmission and reception. The downlink PRS bandwidth may have a granularity of four PRBs, with a minimum of 24 PRBs and a maximum of 272 PRBs. Currently, up to four frequency layers have been defined, and up to two PRS resource sets may be configured per TRP per frequency layer.

[0108] The concept of a frequency layer is somewhat like the concept of component carriers and bandwidth parts (BWPs), but different in that component carriers and BWPs are used by one base station (or a macro cell base station and a small cell base station) to transmit data channels, while frequency layers are used by several (usually three or more) base stations to transmit PRS. A UE may indicate the number of frequency layers it can support when it sends the

network its positioning capabilities, such as during an LTE positioning protocol (LPP) session. For example, a UE may indicate whether it can support one or four positioning frequency layers.

[0109] FIG. 4B illustrates an example of various channels within a downlink slot of a radio frame. In NR, the channel bandwidth, or system bandwidth, is divided into multiple BWPs. A BWP is a contiguous set of PRBs selected from a contiguous subset of the common RBs for a given numerology on a given carrier. Generally, a maximum of four BWPs can be specified in the downlink and uplink. That is, a UE can be configured with up to four BWPs on the downlink, and up to four BWPs on the uplink. Only one BWP (uplink or downlink) may be active at a given time, meaning the UE may only receive or transmit over one BWP at a time. On the downlink, the bandwidth of each BWP should be equal to or greater than the bandwidth of the SSB, but it may or may not contain the SSB.

[0110] Referring to FIG. 4B, a primary synchronization signal (PSS) is used by a UE to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a PCI. Based on the PCI, the UE can determine the locations of the aforementioned DL-RS. The physical broadcast channel (PBCH), which carries an MIB, may be logically grouped with the PSS and SSS to form an SSB (also referred to as an SS/PBCH). The MIB provides a number of RBs in the downlink system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH, such as system information blocks (SIBs), and paging messages.

[0111] The physical downlink control channel (PDCCH) carries downlink control information (DCI) within one or more control channel elements (CCEs), each CCE including one or more RE group (REG) bundles (which may span multiple symbols in the time domain), each REG bundle including one or more REGs, each REG corresponding to 12 resource elements (one resource block) in the frequency domain and one OFDM symbol in the time domain. The set of physical resources used to carry the PDCCH/DCI is referred to in NR as the control resource set (CORESET). In NR, a PDCCH is confined to a single CORESET and is transmitted with its own DMRS. This enables UE-specific beamforming for the PDCCH.

[0112] In the example of FIG. 4B, there is one CORESET per BWP, and the CORESET spans three symbols (although it may be only one or two symbols) in the time domain. Unlike LTE control channels, which occupy the entire system bandwidth, in NR, PDCCH channels are localized to a specific region in the frequency domain (i.e., a CORESET). Thus, the frequency component of the PDCCH shown in FIG. 4B is illustrated as less than a single BWP in the frequency domain. Note that although the illustrated CORESET is contiguous in the frequency domain, it need not be. In addition, the CORESET may span less than three symbols in the time domain.

[0113] The DCI within the PDCCH carries information about uplink resource allocation (persistent and non-persistent) and descriptions about downlink data transmitted to the UE, referred to as uplink and downlink grants, respectively.

More specifically, the DCI indicates the resources scheduled for the downlink data channel (e.g., PDSCH) and the uplink data channel (e.g., PUSCH). Multiple (e.g., up to eight) DCIs can be configured in the PDCCH, and these DCIs can have one of multiple formats. For example, there are different DCI formats for uplink scheduling, for downlink scheduling, for uplink transmit power control (TPC), etc. A PDCCH may be transported by 1, 2, 4, 8, or 16 CCEs in order to accommodate different DCI payload sizes or coding rates.

[0114] Note that the terms “positioning reference signal” and “PRS” generally refer to specific reference signals that are used for positioning in NR and LTE systems. However, as used herein, the terms “positioning reference signal” and “PRS” may also refer to any type of reference signal that can be used for positioning, such as but not limited to, PRS as defined in LTE and NR, TRS, PTRS, CRS, CSI-RS, DMRS, PSS, SSS, SSB, SRS, UL-PRS, etc. In addition, the terms “positioning reference signal” and “PRS” may refer to downlink or uplink positioning reference signals, unless otherwise indicated by the context. If needed to further distinguish the type of PRS, a downlink positioning reference signal may be referred to as a “DL-PRS,” and an uplink positioning reference signal (e.g., an SRS-for-positioning, PTRS) may be referred to as an “UL-PRS.” In addition, for signals that may be transmitted in both the uplink and downlink (e.g., DMRS, PTRS), the signals may be prepended with “UL” or “DL” to distinguish the direction. For example, “UL-DMRS” may be differentiated from “DL-DMRS.”

[0115] With further reference to DL-PRS, DL-PRS have been defined for NR positioning to enable UEs to detect and measure more neighboring TRPs. Several configurations are supported to enable a variety of deployments (e.g., indoor, outdoor, sub-6 GHz, mmW). In addition, beam sweeping is supported for PRS to support PRS beam operation. The following table illustrates various types of reference signals that can be used for various positioning methods supported in NR.

TABLE 1

DL/UL Reference Signals	UE Measurements	To support the following positioning techniques
DL-PRS	DL-RSTD	DL-TDOA
DL-PRS	DL-PRS RSRP	DL-TDOA, DL-AoD, Multi-RTT
DL-PRS/SRS-for-positioning	UE Rx-Tx	Multi-RTT
SSB/CSI-RS for RRM	Synchronization Signal (SS)-RSRP (RSRP for RRM), SS-RSRQ (for RRM), CSI-RSRP (for RRM), CSI-RSRQ (for RRM)	E-CID

[0116] As noted above, NR supports various DL-PRS resource repetition and beam sweeping options. There are several purposes for the repetition of a DL-PRS resource, including (1) receive beam sweeping across repetitions, (2) combining gains for coverage extension, and (3) intra-instance muting. The following table shows the parameters to configure PRS repetition.

TABLE 2

Parameter	Functionality
PRS-ResourceRepetitionFactor	Number of times each PRS resource is repeated for a single instance of the PRS resource set. Values: 1, 2, 4, 6, 8, 16, 32
PRS-ResourceTimeGap	Offset in units of slots between two repeated instances of a DL-PRS resource corresponding to the same PRS resource ID within a single instance of the DL-PRS resource set. Values: 1, 2, 4, 8, 16, 32

[0117] FIG. 5 is a diagram of example PRS resource sets having different time gaps, according to aspects of the disclosure. In the example of FIG. 5, time is represented horizontally and frequency is represented vertically. Each block represents a slot in the time domain and some bandwidth in the frequency domain.

[0118] FIG. 5 illustrates two DL-PRS resource set configurations, a first DL-PRS resource set configuration 510 and a second DL-PRS resource set configuration 550. Each DL-PRS resource set configuration 510 and 550 comprises four PRS resources (labeled “Resource 1,” “Resource 2,” “Resource 3,” and “Resource 4”) and has a repetition factor of four. A repetition factor of four means that each of the four PRS resources is repeated four times (i.e., is transmitted four times) within the DL-PRS resource set. That is, there are four repetitions of each of the four PRS resources within the DL-PRS resource set.

[0119] The DL-PRS resource set configuration 510 has a time gap of one slot, meaning that each repetition of a PRS resource (e.g., “Resource 1”) starts on the first slot after the previous repetition of that PRS resource. Thus, as illustrated by DL-PRS resource set configuration 510, the four repetitions of each of the four PRS resources are grouped together.

[0120] Specifically, the four repetitions of PRS resource “Resource 1” occupy the first four slots (i.e., slots n to $n+3$) of the DL-PRS resource set configuration 510, the four repetitions of PRS resource “Resource 2” occupy the second four slots (i.e., slots $n+4$ to $n+7$), the four repetitions of PRS resource “Resource 3” occupy the third four slots (i.e., slots $n+8$ to $n+11$), and the four repetitions of PRS resource “Resource 4” occupy the last four slots (i.e., slots $n+12$ to $n+15$).

[0121] In contrast, the DL-PRS resource set configuration 550 has a time gap of four slots, meaning that each repetition of a PRS resource (e.g., “Resource 2”) starts on the fourth slot after the previous repetition of that PRS resource. Thus, as illustrated by DL-PRS resource set configuration 550, the four repetitions of each of the four PRS resources are scheduled every fourth slot. For example, the four repetitions of PRS resource “Resource 1” occupy the first, fifth, ninth, and thirteenth slots (i.e., slots n , $n+4$, $n+8$, and $n+12$) of the DL-PRS resource set configuration 550.

[0122] Note that the time duration spanned by one DL-PRS resource set containing repeated DL-PRS resources, as illustrated in FIG. 5, should not exceed the PRS periodicity. In addition, UE receive beam sweeping, for receiving/measuring the DL-PRS resource set, is not specified, but rather, depends on UL implementation.

[0123] A UE has various DL-PRS processing and buffering capabilities that need to be accommodated. For example, a limit on the maximum number of DL-PRS resources

configured to the UL for all TRPs within a measurement window may be defined. In addition, the duration of DL-PRS symbols in units of milliseconds (ms) that a UE can process every T ms assuming some maximum PRS bandwidth, may be defined. The following table shows various parameters indicating a UE's capabilities.

TABLE 3

Description	Maximum Number of DL-PRS Resources	Values that can be Signaled as the UE Capability
Number of frequency layers (X1)	X1 = 4	Values = {1, 4}
Number of TRPs per frequency layer (X2)	X2 = 64	
Number of PRS resource sets per TRP (X3) per frequency layer	X3 = 2	Values = {1, 2}
Number of PRS resources per PRS resource set (X4)	X4 = 64	
Number of DL-PRS resources per UE (X5)	NA	Values = {1, 2}
Number of TRPs for all frequency layers (X6) per UE	X6 = 256	
Number of PRS resources per frequency layer (X7)	NA	

[0124] The following table shows various differences between PRS in LTE and NR.

TABLE 4

Parameter	LTE PRS Baseline	NR PRS
Number of PRS resources per TRP	Each TRP can configure one PRS resource every T ms	Each TRP can configure X PRS resources every T ms. Example values: X = 64 for FR2, X = 8 for FR1 time division duplex (TDD)
FFT Size	2K	4K
Number of OFDM symbols with PRS per PRS resource	8 symbols per subframe with repetition over 6 subframes	Up to 12 symbols per slot with repetition of 32 slots

[0125] A UE reports the duration 'N' of DL-PRS symbols in units of milliseconds that the UE can process every 'T' ms (referred to as a PRS processing window), with the assumption that the UE is capable of processing the maximum PRS bandwidth (e.g., 272 PRBs) in the frequency domain. In addition, a limit on the maximum number of DL-PRS resources configured to the UE for all TRPs within a measurement window is defined. This limit can also be signaled as a UE capability.

[0126] The reporting granularity for UE/gNB timing measurements (e.g., DL-RSTD, UE Rx-Tx time difference, UL-RTOA, gNB Tx-Rx time difference, etc.) is defined as $T = T_c 2^k$, where k is a configuration parameter with a minimum value of at most '0.' Referring to measurement report mapping for RSTD positioning, FIG. 6 illustrates various information elements (IEs) for reporting RSTD measurements. Specifically, an "NR-DL-TDOA-MeasElement-r16" 610 includes 0 to 'X' RSTD measurements in 0 to 'X' "nr-RSTD-r16" fields. The "NR-DL-TDOA-MeasElement-r16" 610 also includes an "nr-DL-TDOA-AdditionalMeasurements-r16" field pointing to an "NR-DL-TDOA-Addi-

tionalMeasurements-r16" IE 620. The "NR-DL-TDOA-AdditionalMeasurements-r16" IE 620 includes a sequence of "NR-DL-TDOA-AdditionalMeasurementElement-r16" fields, each pointing to an "NR-DL-TDOA-AdditionalMeasurementElement-r16" IE 630.

[0127] A parameter 'k' is used to represent the reporting granularity for RSTD measurements, Rx-Tx time difference measurements, etc. The location server (e.g., LMF 270) provides a recommended value of 'k', referred to as 'k1' (also denoted 'k₁'). The UE then selects a value of 'k', referred to as 'k2' (also denoted 'k₂'), and informs the location server of the selected value. The relation between the UE-selected parameter 'k2' and the network-recommended value 'k1' needs to be determined. In addition, for RSTD and UE Rx-Tx time difference reporting, the relationship between 'k1' and 'k2' and the range of 'k1' and 'k2' in FR1 need to be determined.

[0128] The location server should set the value of 'k1' properly with respect to the PRS bandwidth. That is, with increasing PRS bandwidth, 'k1' should be set to smaller values and vice versa. With this understanding, the value that the UE selects, 'k2,' can be equal to, or larger than 'k1.' In most scenarios, the UE should adhere to the recommended value from the location server (i.e., k1=k2), but in cases where the PRS bandwidth is larger than the active downlink BWP and a proper measurement gap (a period of time during which the UE's serving base station does not communicate with the UE to allow the UE to measure PRS from other base stations) has not been configured or not yet requested by the UE, then 'k1' may be less than or equal to 'k2.'

[0129] Another consideration is that the UE is expected to meet the accuracy requirements for each PRS bandwidth as long as it supports that PRS bandwidth. Thus, the UE needs to use a 'k2' value that satisfies

$$k_2 \leq \left\lceil \log_2 \left(\frac{1}{4T_c BW_{PRS}} \right) \right\rceil.$$

[0130] To account for possible differences between the bandwidth of the reference PRS resource or resource set (e.g., the PRS resource(s) or resource set(s) from the reference TRP for an RSTD measurement) and neighbor PRS resource or resource set (e.g., the PRS resource(s) or resource set(s) from neighbor TRPs for an RSTD measurement), a minimum between the two bandwidths should be taken. Moreover, the UE may be configured to report, subject to UE capability, up to four DL-RSTD measurements per pair of TRPs with each measurement between a different pair of DL-PRS resources or DL-PRS resource sets within the DL-PRS configured for those TRPs. This is shown in FIG. 6, where there may be up to three "NR-DL-TDOA-AdditionalMeasurementElement-r16" fields in the "NR-DL-TDOA-AdditionalMeasurements-r16" IE 620 in addition to the measurement reported in the "NR-DL-TDOA-MeasElement-r16" IE 610. The up to four measurements being performed on the same pair of TRPs and all DL-RSTD measurements in the same report use a single reference timing.

[0131] These (up to four) DL-RSTD measurements on the same TRP pair can belong to different positioning frequency layers, hence the constraints should be based on the minimum PRS bandwidth across positioning frequency layers of

the same TRP pair. Accordingly, the present disclosure proposes the relationship between the location server-recommended value of 'k' (i.e., 'k1') and the UE-selected value of 'k' (i.e., 'k2') to be:

$$k_1 \leq k_2 \leq \left\lceil \log_2 \left(\frac{1}{M \cdot T_c \min_i (BW_{PRS,ref}, BW_{PRS,neighbor,i})} \right) \right\rceil$$

where $BW_{PRS,ref}$ is the bandwidth of the PRS resource or resource set from the reference TRP, $BW_{PRS,neighbor}$ is the bandwidth of the PRS resource or resource set from the neighbor TRP i , i is the positioning frequency layer index, $T_c=0.5$ ns, and M is an oversampling factor. Specifically, M is a constant for all measurements, but it can also be configurable, or the UE may recommend the value. M may depend on the band, FR1/FR2 differentiation, and/or UE capability. For example, M may be selected from the set of {1, 2, 4, 8}. As M increases, the bound on 'k2' decreases, thereby increasing granularity.

[0132] For example, given the above, a UE may measure a first RSTD (denoted "RSTD1") as "t2-t1," a second RSTD (denoted "RSTD2") as "t3-t1," a third RSTD (denoted "RSTD3") as "t4-t1," and a fourth RSTD (denoted "RSTD4") as "t5-t1," where 't1,' 't2,' 't3,' 't4,' and 't5' are the times of arrival (ToAs) of the PRS from the corresponding TRPs. In this example, 't2,' 't3,' and 't4' may be in a different frequency layer than 't1' and 't5,' and 't1' and 't5' may be in the same frequency layer. In this case, all RSTD measurements would use the same value of 'k2.'

[0133] Referring now to measurement report mapping for Rx-Tx positioning, FIGS. 7 and 8 illustrate various information elements (IEs) for reporting Rx-Tx time difference measurements. Specifically, an "NR-Multi-RTT-MeasElement-r16" **710** includes some number of UE Rx-Tx time difference measurements in one or more "nr-UE-RxTxTimeDiff-r16" fields. The "NR-Multi-RTT-MeasElement-r16" **710** also includes an "nr-Multi-RTT-AdditionalMeasurements-r16" field pointing to an "NR-Multi-RTT-AdditionalMeasurements-r16" IE **720**. The "NR-Multi-RTT-AdditionalMeasurements-r16" IE **720** includes a sequence of up to three "NR-Multi-RTT-AdditionalMeasurementElement-r16" fields, each pointing to an "NR-Multi-RTT-AdditionalMeasurementElement-r16" IE **730**. In FIG. 8, an "NR-AdditionalPathList-r16" IE **810** can be used to report up to two additional paths for a UE Rx-Tx time difference measurement. The "NR-AdditionalPath-r16" fields in the "NR-AdditionalPathList-r16" IE **810** point to "NR-AdditionalPath-r16" IEs **820** for reporting the actual measurements. The relationship between the value of 'k' configured by the location server ('k1') and the value of 'k' selected by the UE ('k2') discussed above for RSTD measurements also applies to UE Rx-Tx time difference measurement reporting as well.

[0134] To account for possible differences between the bandwidth of PRS resources for multiple Rx-Tx time difference measurements of the same TRP pair, a minimum between the bandwidth across frequency layers should be taken. Accordingly, the present disclosure proposes the

relationship between the location server-recommended value of 'k' ('k1') and the UE-selected value of 'k' ('k2') to be:

$$k_1 \leq k_2 \leq \left\lceil \log_2 \left(\frac{1}{M \cdot T_c \min_i (BW_{PRS,i})} \right) \right\rceil$$

where BW_{PRS} is the bandwidth of the PRS resource or resource set, i is the positioning frequency layer index, and $T_c=0.5$ ns. The value of $T_c \min_i (BW_{PRS,i})$ is effectively the sampling time, and M is an oversampling factor. M should be a constant for all measurements, but it can also be configurable, or the UE may recommend the value. It may also, or alternatively, depend on the band or FR1/FR2 differentiation.

[0135] Referring now to determining the measurement period for PRS measurements, for the purpose of DL-PRS processing capability, the duration of DL-PRS symbols (K) in ms within any P ms window, is determined by a Type 1 duration calculation or a Type 2 duration calculation. Type 1 or Type 2 is reported as a UE capability.

[0136] The Type 1 duration calculation is:

$$K = \sum_{s \in S} K_s$$

$$K_s = T_s^{end} - T_s^{start}$$

[0137] The Type 2 duration calculation is:

$$K = \frac{1}{2\mu} |S|$$

[0138] In the equations above, S is the set of slots of a serving cell within the P ms window in the positioning frequency layer that contains potential DL-PRS resources considering the actual expected RSTD (e.g., given by the parameter "nr-DL-PRS-ExpectedRSTD"), the actual expected RSTD uncertainty (e.g., given by the parameter "nr-DL-PRS-ExpectedRSTD-Uncertainty") provided for each pair of DL-PRS resources or resource sets (target and reference). For Type 1, $[T_s^{start}, T_s^{end}]$ is the smallest interval in milliseconds within slot s corresponding to an integer number of OFDM symbols of a serving cell that covers the union of the potential PRS symbols and determines the PRS symbol occupancy within slot s . The interval $[T_s^{start}, T_s^{end}]$ considers the actual "nr-DL-PRS-ExpectedRSTD," "nr-DL-PRS-ExpectedRSTD-Uncertainty" provided for each pair of DL-PRS resources or resource sets (target and reference).

[0139] For the UE DL-PRS processing capability, the UE reports one combination of (N , T) values per band, where, as noted above, ' N ' is a duration of DL-PRS symbols in ms the UE can process every ' T ' ms for a given maximum bandwidth ' B ' in MHz supported by the UE. Additionally, the UE reports a new parameter, the number of DL-PRS resources that the UE can process in a slot, which is reported per SCS per band. The reported number of DL-PRS resources that the UE can process in a slot may be selected from the set of {1, 2, 4, 8, 12, 16, 32, 64}.

[0140] The following sets of values for 'N,' 'T,' and 'B' are currently supported. Values for 'N' may be selected from the set of {0.125, 0.25, 0.5, 1, 2, 4, 8, 12, 16, 20, 25, 30, 35, 40, 45, 50} ms., values for 'T' may be selected from the set of {8, 16, 20, 30, 40, 80, 160, 320, 640, 1280} ms, and values for the maximum bandwidth reported by the UE ('B') may be selected from the set of {5, 10, 20, 40, 50, 80, 100, 200, 400} MHz.

[0141] The reporting of (N, T) values for the maximum bandwidth in MHz is not dependent on the SCS. The UE capability for simultaneous DL-PRS processing across positioning frequency layers is not currently supported (i.e., for a UE supporting multiple positioning frequency layers, a UE is expected to process one frequency layer at a time). The UE capability for DL-PRS processing is defined assuming the case with configured measurement gap and a maximum ratio of measurement gap length (MGL)/measurement gap repetition period (MGRP) of no more than 'X'%.

[0142] As described above, two categories are defined for the PRS duration calculation: Type 1 and Type 2. In Type 2, the presence of even one PRS resource within a slot results in the entire slot duration being considered for PRS duration calculation regardless of how many symbols it occupies. Compared to Type 1, the Type 2 duration calculation is more conservative, but the two types are approximately the same when PRS symbols occupy most of a slot duration. For the purpose of measurement duration, using only the more conservative Type 2 would be preferable over having two separate formulations for Type 1 and Type 2.

[0143] Accordingly, the present disclosure proposes to define the RSTD measurement period based on the Type 2 PRS duration calculation. As such, the RSTD measurement period for a Type 1 PRS duration calculation should be no longer than Type 2.

[0144] To arrive at a basic formulation of the measurement period, the followings are assumed: (1) one positioning frequency layer, (2) one common PRS periodicity among all configured PRS resources, (3) no loss of PRS occasions due to handover (HO), overlap with SSB, or other reasons, (4) no receive beam sweeping (FR1), and (5) one sample per PRS resource.

[0145] FIG. 9 is a diagram 900 of several DL-PRS resources spanning a given time duration in milliseconds, according to aspects of the disclosure. In the example of FIG. 9, time is represented horizontally and frequency is represented vertically. Each block represents a symbol in the time domain and some amount of bandwidth in the frequency domain.

[0146] In the example of FIG. 9, there are three DL-PRS resources (differentiated by different hashing) that each have a repetition factor of four within a slot (14 symbols). The DL-PRS resources may be part of the same or different DL-PRS resource sets. A repetition factor of four means that each of the three PRS resources is repeated four times (i.e., is transmitted four times) within a slot. The DL-PRS resources have a time gap of one symbol, meaning that each repetition of a PRS resource starts on the first symbol after the previous repetition of that PRS resource. Thus, as illustrated in FIG. 9, the four repetitions of each of the three PRS resources are grouped together in four consecutive symbols.

[0147] In the example of FIG. 9, the first two slots (i.e., the first two groups of 14 symbols) correspond to the first and last slots of a PRS occasion or instance. The length of the

PRS occasion is denoted L_{PRS} (labeled "L_PRS"). Specifically, L_{PRS} is the time from the first symbol of the first DL-PRS resource of the PRS occasion to the last symbol of the last DL-PRS resource of the PRS occasion. As such, in the example of FIG. 9, L_{PRS} spans L_{PRS} slots, that is, from slot '0' to slot ' $L_{PRS}-1$ ' (labeled "L_PRS-1").

[0148] As shown in FIG. 9, the PRS periodicity, that is, the time from the first repetition of the first DL-PRS resource of a first PRS instance to the same first repetition of the same first DL-PRS resource of the next PRS instance, is represented as T_{PRS} (labeled "T_PRS"). As such, in the example of FIG. 9, the PRS occasion is illustrated as spanning T_{PRS} slots, that is, from slot '0' to slot ' $T_{PRS}-1$ ' (labeled "T_PRS-1").

[0149] A UE may measure, or sample, one or more repetitions of a PRS resource in a slot, depending on the UE's reported capabilities for 'N' (the duration of DL-PRS symbols in milliseconds the UE can process every 'T' ms) and 'T' (the number of milliseconds within which the UE can process a duration 'N' ms of DL-PRS). Such measurements or samples may also be referred to as measurement instances. If L_{PRS} is less than or equal to 'N,' then a UE only needs 'T' ms to process the PRS resources. Otherwise, the UE needs to measure PRS resources in a round-robin fashion, and the measurement period is scaled by the maximum of 'T,' the measurement gap period (MGP), and T_{PRS} . Similarly, if the number of PRS resources in a slot (denoted N_{PRS}^{slot}) is less than or equal to 'N' prime (N'), then the UE only needs 'T' ms to process the PRS resources. Otherwise, the measurement period is scaled similarly as in the 'N' case.

[0150] Accordingly, the present disclosure defines the RSTD measurement period as:

$$T_{RSTD,base} = \left\{ \max \left(\left\lceil \frac{L_{PRS}}{N} \right\rceil, \left\lceil \frac{N_{PRS}^{slot}}{N'} \right\rceil \right) - 1 \right\} \cdot \max(T, T_{PRS}, MGP) + T$$

where 'N' is a duration of DL-PRS symbols in milliseconds the UE can process every 'T' ms for a given maximum bandwidth 'B' in MHz supported by the UE, 'N' prime (N') is the number of DL-PRS resources that the UE can process in a slot, which is reported per SCS per band, L_{PRS} represents the span of a PRS occasion (or instance) defined as the time from the first slot of the earliest PRS resource to the last slot of latest PRS resource based on Type 2 duration calculation, N_{PRS}^{slot} is the number of configured PRS resources in a slot, T_{PRS} is the periodicity of PRS, and MGP is the measurement gap period. The parameter

$$\left(\left\lceil \frac{L_{PRS}}{N} \right\rceil, \left\lceil \frac{N_{PRS}^{slot}}{N'} \right\rceil \right)$$

defines the number of measurement instances and the parameter $\max(T, T_{PRS}, MGP)$ is a bandwidth periodicity parameter.

[0151] FIG. 10 illustrates an example method 1000 of wireless communication, according to aspects of the disclosure. In an aspect, method 1000 may be performed by a UE (e.g., any of the UEs described herein).

[0152] At 1010, the UE receives, from a network entity (e.g., LMF 270), a configuration of one or more PRS resources to measure during a positioning session, the one or

more PRS resources having a PRS periodicity T_{PRS} and a PRS occasion length L_{PRS} . In an aspect, operation **1010** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

[0153] At **1020**, the UE measures the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of 'T' milliseconds, the PRS periodicity T_{PRS} , and a measurement gap periodicity of at least one measurement gap. In an aspect, operation **1020** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

[0154] FIG. **11** illustrates an example method **1100** of wireless communication, according to aspects of the disclosure. In an aspect, method **1100** may be performed by a UE (e.g., any of the UEs described herein).

[0155] At **1110**, the UE receives, from a network entity (e.g., LMF **270**), a recommendation of a first granularity (e.g., 'k1') for reporting positioning measurements of one or PRS resources during a positioning session. In an aspect, operation **1110** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

[0156] At **1120**, the UE performs one or more positioning measurements of the one or more PRS resources. In an aspect, operation **1120** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

[0157] At **1130**, the UE reports the one or more positioning measurements at a second granularity ('k2'), wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources. In an aspect, operation **1130** may be performed by the one or more WWAN transceivers **310**, the one or more processors **332**, memory **340**, and/or positioning component **342**, any or all of which may be considered means for performing this operation.

[0158] As will be appreciated, a technical advantage of the method **1000** and **1100** is enabling communication between the location server and a UE as to how long a measurement period the UE needs, thereby enabling the location server and/or the UE to determine the latency of a positioning procedure and/or positioning measurements.

[0159] In the detailed description above it can be seen that different features are grouped together in examples. This manner of disclosure should not be understood as an intention that the example clauses have more features than are explicitly mentioned in each clause. Rather, the various aspects of the disclosure may include fewer than all features of an individual example clause disclosed. Therefore, the following clauses should hereby be deemed to be incorpo-

rated in the description, wherein each clause by itself can stand as a separate example. Although each dependent clause can refer in the clauses to a specific combination with one of the other clauses, the aspect(s) of that dependent clause are not limited to the specific combination. It will be appreciated that other example clauses can also include a combination of the dependent clause aspect(s) with the subject matter of any other dependent clause or independent clause or a combination of any feature with other dependent and independent clauses. The various aspects disclosed herein expressly include these combinations, unless it is explicitly expressed or can be readily inferred that a specific combination is not intended (e.g., contradictory aspects, such as defining an element as both an insulator and a conductor). Furthermore, it is also intended that aspects of a clause can be included in any other independent clause, even if the clause is not directly dependent on the independent clause.

[0160] Implementation examples are described in the following numbered clauses:

[0161] Clause 1. A method of wireless positioning performed by a user equipment (UE), comprising: receiving, from a network entity, a configuration of one or more positioning reference signal (PRS) resources to measure during a positioning session, the one or more PRS resources having a PRS periodicity T_{PRS} and a PRS occasion length L_{PRS} ; and measuring the one or more PRS resources during a measurement period, wherein the measurement period is based on a number of measurement instances of the one or more PRS resources the UE is expected to process multiplied by a periodicity parameter, wherein the periodicity parameter is based on a PRS processing window of 'T' milliseconds, the PRS periodicity T_{PRS} , and a measurement gap periodicity of at least one measurement gap.

[0162] Clause 2. The method of clause 1, wherein the periodicity parameter is a maximum of the PRS processing window, the PRS periodicity T_{PRS} , and the measurement gap periodicity.

[0163] Clause 3. The method of any of clauses 1 to 2, wherein the periodicity parameter is no less than a maximum of the PRS processing window, the PRS periodicity T_{PRS} , and the measurement gap periodicity.

[0164] Clause 4. The method of any of clauses 1 to 3, wherein the number of measurement instances is based on the PRS occasion length L_{PRS} .

[0165] Clause 5. The method of clause 4, wherein the PRS occasion length L_{PRS} is based on whether a Type 1 duration calculation or a Type 2 duration calculation is used to determine a duration of PRS symbols in milliseconds within any PRS processing window.

[0166] Clause 6. The method of any of clauses 1 to 5, wherein 'T' milliseconds is number of milliseconds during which the UE can process a duration 'N' of PRS symbols in milliseconds given a maximum PRS bandwidth.

[0167] Clause 7. The method of any of clauses 1 to 6, wherein the PRS occasion length L_{PRS} is a time from a first symbol of a first PRS resource of a PRS occasion to a last symbol of a last PRS resource of the PRS occasion.

[0168] Clause 8. The method of any of clauses 1 to 7, wherein the PRS periodicity T_{PRS} is a time from a first repetition of a first PRS resource of a first PRS occasion to the first repetition of the first PRS resource of a next PRS occasion.

[0169] Clause 9. The method of any of clauses 1 to 8, wherein: the measurement period is a reference signal time difference (RSTD) measurement period, and the one or more PRS resources are transmitted by a reference transmission-reception point (TRP) and at least one neighbor TRP.

[0170] Clause 10. The method of any of clauses 1 to 8, wherein the measurement period is a reception-to-transmission (Rx-Tx) time difference measurement period.

[0171] Clause 11. A method of wireless positioning performed by a user equipment (UE), comprising: receiving, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session; performing one or more positioning measurements of the one or more PRS resources; and reporting the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

[0172] Clause 12. The method of clause 11, wherein the PRS bandwidth parameter includes an inverse of a minimum bandwidth of the one or more PRS resources.

[0173] Clause 13. The method of any of clauses 11 to 12, wherein the PRS bandwidth parameter includes a minimum bandwidth of the one or more PRS resources across all positioning frequency layers on which the one or more PRS resources are transmitted.

[0174] Clause 14. The method of any of clauses 11 to 13, wherein the PRS bandwidth parameter includes an inverse of an oversampling factor.

[0175] Clause 15. The method of clause 14, wherein the oversampling factor: is configured to the UE by the network entity, is recommended by the UE, is based on a frequency range of the one or more PRS resources, or is a capability of the UE.

[0176] Clause 16. The method of any of clauses 11 to 15, wherein the PRS bandwidth parameter includes a minimum bandwidth of the one or more PRS resources across a reference PRS resource of the one or more PRS resources and at least one neighbor PRS resource of the one or more PRS resources.

[0177] Clause 17. The method of any of clauses 15 to 16, wherein the reference PRS resource is transmitted by a reference transmission-reception point (TRP) and the at least one neighbor PRS resource is transmitted by at least one neighbor TRP.

[0178] Clause 18. The method of any of clauses 11 to 17, wherein the PRS bandwidth parameter includes a logarithmic function of an inverse of a bandwidth of the one or more PRS resources.

[0179] Clause 19. The method of any of clauses 11 to 18, wherein the one or more positioning measurements comprise one or more reference signal time difference (RSTD) measurements.

[0180] Clause 20. The method of any of clauses 11 to 19, wherein the one or more positioning measurements comprise one or more reception-to-transmission (Rx-Tx) time difference measurements.

[0181] Clause 21. An apparatus comprising a memory, at least one transceiver, and at least one processor communicatively coupled to the memory and the at least one transceiver, the memory, the at least one transceiver, and the at

least one processor configured to perform a method according to any of clauses 1 to 20.

[0182] Clause 22. An apparatus comprising means for performing a method according to any of clauses 1 to 20.

[0183] Clause 23. A non-transitory computer-readable medium storing computer-executable instructions, the computer-executable comprising at least one instruction for causing a computer or processor to perform a method according to any of clauses 1 to 20.

[0184] Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

[0185] Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

[0186] The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an ASIC, a field-programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0187] The methods, sequences and/or algorithms described in connection with the aspects disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An example storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal (e.g.,

UE). In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

[0188] In one or more example aspects, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0189] While the foregoing disclosure shows illustrative aspects of the disclosure, it should be noted that various changes and modifications could be made herein without departing from the scope of the disclosure as defined by the appended claims. The functions, steps and/or actions of the method claims in accordance with the aspects of the disclosure described herein need not be performed in any particular order. Furthermore, although elements of the disclosure may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated.

What is claimed is:

1. A method of wireless positioning performed by a user equipment (UE), comprising:

receiving, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session;

performing one or more positioning measurements of the one or more PRS resources; and

reporting the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

2. The method of claim 1, wherein the PRS bandwidth parameter includes an inverse of a minimum bandwidth of the one or more PRS resources.

3. The method of claim 1, wherein the PRS bandwidth parameter includes a minimum bandwidth of the one or more PRS resources across all positioning frequency layers on which the one or more PRS resources are transmitted.

4. The method of claim 1, wherein the PRS bandwidth parameter includes an inverse of an oversampling factor.

5. The method of claim 4, wherein the oversampling factor:

is configured to the UE by the network entity,

is recommended by the UE,

is based on a frequency range of the one or more PRS resources, or

is a capability of the UE.

6. The method of claim 1, wherein the PRS bandwidth parameter includes a minimum bandwidth of the one or more PRS resources across a reference PRS resource of the one or more PRS resources and at least one neighbor PRS resource of the one or more PRS resources.

7. The method of claim 6, wherein the reference PRS resource is transmitted by a reference transmission-reception point (TRP) and the at least one neighbor PRS resource is transmitted by at least one neighbor TRP.

8. The method of claim 1, wherein the PRS bandwidth parameter includes a logarithmic function of an inverse of a bandwidth of the one or more PRS resources.

9. The method of claim 1, wherein the one or more positioning measurements comprise one or more reference signal time difference (RSTD) measurements.

10. The method of claim 1, wherein the one or more positioning measurements comprise one or more reception-to-transmission (Rx-Tx) time difference measurements.

11. A user equipment (UE), comprising:

a memory;

at least one transceiver; and

at least one processor communicatively coupled to the memory and the at least one transceiver, the at least one processor configured to:

receive, via the at least one transceiver, from a network entity, a recommendation of a first granularity for reporting positioning measurements of one or more positioning reference signal (PRS) resources during a positioning session;

perform one or more positioning measurements of the one or more PRS resources; and

report the one or more positioning measurements at a second granularity, wherein the second granularity is less than or equal to the first granularity and greater than or equal to a minimum granularity, and wherein the minimum granularity is based on a PRS bandwidth parameter associated with the one or more PRS resources.

12. The UE of claim 11, wherein the PRS bandwidth parameter includes an inverse of a minimum bandwidth of the one or more PRS resources.

13. The UE of claim 11, wherein the PRS bandwidth parameter includes a minimum bandwidth of the one or more PRS resources across all positioning frequency layers on which the one or more PRS resources are transmitted.

14. The UE of claim 11, wherein the PRS bandwidth parameter includes an inverse of an oversampling factor.

15. The UE of claim **14**, wherein the oversampling factor: is configured to the UE by the network entity, is recommended by the UE, is based on a frequency range of the one or more PRS resources, or is a capability of the UE.

16. The UE of claim **11**, wherein the PRS bandwidth parameter includes a minimum bandwidth of the one or more PRS resources across a reference PRS resource of the one or more PRS resources and at least one neighbor PRS resource of the one or more PRS resources.

17. The UE of claim **16**, wherein the reference PRS resource is transmitted by a reference transmission-reception point (TRP) and the at least one neighbor PRS resource is transmitted by at least one neighbor TRP.

18. The UE of claim **11**, wherein the PRS bandwidth parameter includes a logarithmic function of an inverse of a bandwidth of the one or more PRS resources.

19. The UE of claim **11**, wherein the one or more positioning measurements comprise one or more reference signal time difference (RSTD) measurements.

20. The UE of claim **11**, wherein the one or more positioning measurements comprise one or more reception-to-transmission (Rx-Tx) time difference measurements.

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